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Sustainability criteria for the blue economy

Annexes to the main report

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European Climate, Infrastructure and Networks Executive Agency (CINEA)
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European Climate, Infrastructure and Networks Executive Agency (CINEA)

European Maritime and Fisheries Fund

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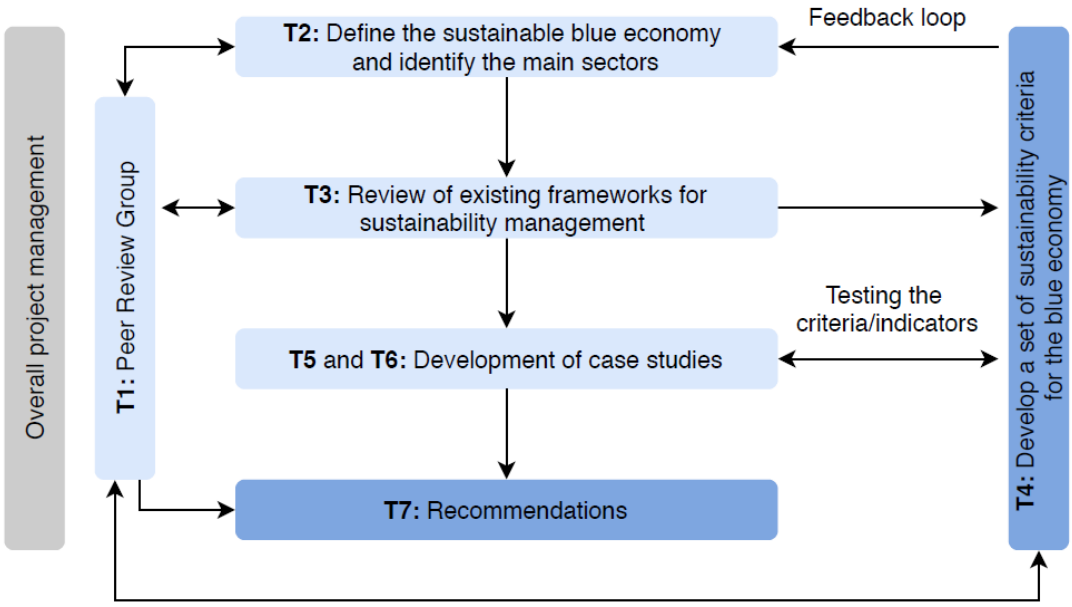
Introduction

The annex document is submitted as an attachment to the final report for the Blue Economy Framework – Lot 2 – Sustainability Criteria for a Blue Economy. It provides the reader with an in-depth overview of the methodologies applied to each task.

1. Task 2: is covered by the annexes 1 to 3.
2. Task 3 is covered by the annexes 4 to 6.
3. Task 4 is covered by the annex 7, 8, 9 and 10.

Figure 1: Overview of research tasks



Annex 1: Review of existing definitions for a blue economy

The following table displays the 32 existing definitions of a blue economy that have been taken into account for Task 2.

Table 1: Sources of blue economy definitions

Organisation	Year	Definitions	Reference
European Commission	2012	All economic activities that depend on the sea. The individual sectors of the Blue Economy are interdependent. They rely on common skills and shared infrastructure such as ports and electricity distribution networks. They depend on others using the sea sustainably.	European Commission (2012). Blue Growth – opportunities for marine and maritime sustainable growth. European Commission Publication Office, Brussels, Belgium
Changwon Declaration	2012	We understand the Blue Economy to be a practical ocean-based economic model using green infrastructure and technologies, innovative financing mechanisms and proactive institutional arrangements for meeting the twin goals of protecting our oceans and coasts and enhancing its potential contribution to sustainable development, including improving human well-being, and reducing environmental risks and ecological scarcities.	Changwon Declaration (2012). Towards an Ocean based economy: Moving ahead with the sustainable development strategy for the seas of east Asia.
Ecorys et al.	2012	Blue Growth is hence defined as "smart, sustainable and inclusive economic and employment growth from the oceans, seas and coasts".	Ecorys, Deltares and Oceanic Développement (2012). Blue Growth—Scenarios and Drivers for Sustainable Growth from the Oceans, Seas and Coasts. Final Report. Commissioned by DG. MARE, Brussels, Belgium.
Government of Australia	2013	A Blue Economy is one in which our ocean ecosystems bring economic and social benefits that are efficient, equitable and sustainable.	Coffin, M. (2013). Marine Nation 2025: Marine Science to Support Australia's Blue Economy.
United States National Ocean Economics Program	2014	The 'Blue Economy' is a widely-used term meaning the overall economic contribution of the oceans and coasts to the national economy and the need to address the sustainability of these natural assets.	US NOEP (2014) Centre for the Blue Economy Retrieved from www.centerfortheblueeconomy.org on 1-11-2019.

Economist Intelligence Unit	2015	A sustainable ocean economy emerges when economic activity is in balance with the long-term capacity of ocean ecosystems to support this activity and remain resilient and healthy.	EIU (2015). The blue economy: Growth, opportunity and a sustainable ocean economy. Briefing Paper Prepared for the World Ocean Summit 2015.
Partnerships in Environmental Management for the Seas of East Asia	2015	Beyond simply being a collection of coastal and marine industries, Blue Economy is the set of environmentally and socially sustainable commercial activities, products, services and investments dependent on and impacting coastal and marine resources.	Whisnant, R. and Reyes, A. (2015). Blue Economy for Business in East Asia: Towards an Integrated Understanding of Blue Economy. Partnerships in Environmental Management for the Seas of East Asia (PEMSEA), Quezon City, Philippines. 69 p.
National Marine Science Committee	2015	A Blue Economy is one which strikes the right balance between reaping the economic potential of our oceans with the need to safeguard their longer term health.	NMSC (2015). National Marine Science Plan 2015-2025: Driving the development of Australia's blue economy.
Research and Information System for Developing Countries	2015a	As a segment of an economy, Ocean economy is dependent on ocean for inputs to run certain production processes in the coastal and non-coastal regions, and these activities are identified by both industry and geographical location.	Mohanty, S. K., Dash, P., Gupta, A. and Gaur, P. (2015). Prospects of Blue Economy in the Indian Ocean. Research and Information System for Developing Countries, New Delhi.
Research and Information System for Developing Countries	2015b	As a sub-set of the economy, Blue Economy covers all ocean related activities including direct and indirect supporting activities required for functioning of these economic sectors, while adjusting to the costs of environmental damage and ecological imbalance caused due to exploitation of ocean resources for consumption.	Mohanty, S. K., Dash, P., Gupta, A., & Gaur, P. (2015). Prospects of Blue Economy in the Indian Ocean. Research and Information System for Developing Countries, New Delhi.
United Nations Environment Programme	2016	A Blue Economy is low-carbon, efficient, and clean. It is also an economy that is based on sharing, circularity, collaboration, solidarity, resilience, opportunity and interdependence. Its growth is driven by investments that reduce carbon emissions and pollution, enhance energy efficiency, harness the power of natural capital - such as the oceans - and halt the loss of biodiversity and the benefits that ecosystems provide.	UNEP (2016). Blue economy. Retrieved from https://www.unenvironment.org/news-and-stories/story/blue-economy on 28-10-2019
OECD	2016	The ocean economy can be defined as the sum of the economic activities of ocean-based industries, and the assets, goods and services of marine ecosystems.	OECD (2016). The Ocean Economy in 2030.OECD Publishing. Paris, France.

Commonwealth Secretariat	2016	One general interpretation of the Blue Economy is that it can maximise the economic value of the marine environment in a sustainable manner that preserves and protects the sea's resources and ecosystems. By that definition, the Blue Economy can be broadly defined to include 'economic activity which directly or indirectly uses the sea as an input.	Roberts, J. and Ali, A. (2016). The Blue Economy and Small States. Commonwealth Blue Economy Series, No. 1. Commonwealth Secretariat, London.
United Nations Economic Commission for Africa	2016	The Blue Economy in the African context covers both aquatic and marine spaces, including oceans, seas, coasts, lakes, rivers, and underground water. It encompasses a range of productive sectors, including fisheries, aquaculture, tourism, transport, shipbuilding, energy, bioprospecting, and underwater mining and related activities.	UNECA (2016). Africa's Blue Economy: A policy handbook. United Nations. Addis Ababa, Ethiopia
Smith-Godfrey	2016	Blue Economy is the sustainable industrialisation of the oceans to the benefit of all.	Smith-Godfrey, S. (2016). Defining the blue economy. Maritime affairs: Journal of the national maritime foundation of India, 12(1), 58-64.
Stats New Zealand	2016	The marine economy is a function of both industry and geography. It is the sum of the economic activities that take place in, or use, the marine environment, or produce goods and services necessary for those activities, and make a direct contribution to the national economy.	Stats New Zealand (2016). Marine economy. Retrieved from http://archive.stats.govt.nz/browse_for_stats/environment/environmental-reporting-series/environmental-indicators/Home/Marine/marine-economy.aspx on 24-10-2019
Ebarvia, M.	2016	The Blue Economy [...] refers to a sustainable ocean-based economic model that is largely dependent on coastal and marine ecosystems and resources, but one that employs environmentally-sound and innovative infrastructure, technologies and practices, including institutional and financing arrangements, for meeting the goals of: (a) sustainable and inclusive development; (b) protecting our coasts and oceans, and reducing environmental risks and ecological scarcities; (c) addressing water, energy and food security; (d) protecting the health, livelihoods and welfare of the people in the coastal zone; and (e) fostering an ecosystem-based climate change mitigation and adaptation measures.	Ebarvia, M. (2016). Economic assessment of oceans for sustainable Blue Economy development. Journal of Ocean and Coastal Economics, 2(2), 7.

European Council	2017	The Blue Economy is a marine-and maritime-based economy which is well-managed, sustainable. It aims at reconciling ocean-related sustainable economic growth with improved livelihoods and social equity for current and future generations and the strengthening of transparent, reliable and more secure food systems, based on the conservation of marine ecosystems and biodiversity and on the sustainable use of resources.	European Commission (2017). Council Conclusions on Blue Growth. Brussels.
Indian Ocean Rim Association	2017	The objective of the Blue Economy is to promote smart, sustainable and inclusive growth and employment opportunities within the Indian Ocean region's maritime economic activities. The Blue Economy is determined to initiate appropriate programs for: the sustainable harnessing of ocean resources; research and development; developing relevant sectors of oceanography; stock assessment of marine resources; introducing marine aquaculture, deep sea/long line fishing and biotechnology; and human resource development; among others.	IORA (2017). Blue Economy. Retrieved from https://www.iora.int/en/priorities-focus-areas/blue-economy on 28-10-2019
Plan Bleu	2017	Blue economy is a low polluting, resource-efficient and circular economy based on sustainable consumption and production patterns, enhancing human well-being and social equity, generating economic value and employment, and significantly reducing environmental risks and ecological scarcities	Giraud, J., Lafitte, A. and Fosse, J. (2017). Blue economy; economic activities and sustainable development. Plan Blue. Notes 34. Valbonne, France.
Food and Agriculture Organization	2017	The blue economy [...] refers to a sustainable ocean-based economic model that is largely dependent on coastal and marine ecosystems and resources, but one that employs environmentally-sound and innovative infrastructure, technologies and practices, including institutional and financing arrangements, for meeting the goals of: (a) sustainable and inclusive development; (b) protecting our coasts and oceans, and reducing environmental risks and ecological scarcities; (c) addressing water, energy and food security; (d) protecting the health, livelihoods and welfare of the people in the coastal zone; and (e) fostering an ecosystem-based climate change mitigation and adaptation measures.	FAO (2017). Blue Growth Initiative. Retrieved from http://www.fao.org/3/a-i7862e.pdf on 1-11-2019.
World Bank	2017a	[..] the term "Blue Economy [..] is understood here as comprising the range of economic sectors and related policies that together determine whether the use of oceanic resources is sustainable.	Vierros, Marjo, and De Fontaubert Charlotte (2017). The potential of the blue economy: increasing long-term benefits of the sustainable use of marine resources for

			small island developing states and coastal least developed countries (English). Washington, D.C. : World Bank Group.
World Bank	2017b	The 'Blue Economy' concept seeks to promote economic growth, social inclusion, and preservation or improvement of livelihoods while at the same time ensuring environmental sustainability. At its core, it refers to the decoupling of socioeconomic development through oceans-related sectors and activities from environmental and ecosystems degradation. It draws from scientific findings that ocean resources are limited and that the health of the oceans has drastically declined due to anthropogenic activities.	World Bank &UN DESA (2017). The Potential of the Blue Economy: Increasing Long-term Benefits of the Sustainable Use of Marine Resources for Small Island Developing States and Coastal Least Developed Countries. World Bank, Washington DC
World Wildlife Fund for Nature	2018	A Sustainable Blue Economy is a marine-based economy that: • Provides social and economic benefits for current and future generations, by contributing to food security, poverty eradication, livelihoods, income, employment, health, safety, equity, and political stability. • Restores, protects and maintains the diversity, productivity, resilience, core functions, and intrinsic value of marine ecosystems - the natural capital upon which its prosperity depends. • Is based on clean technologies, renewable energy, and circular material flows to secure economic and social stability over time, while keeping within the limits of one planet.	WWF (2018). Principles for a Sustainable BLUE ECONOMY. Retrieved from https://d2ouvy59p0dg6k.cloudfront.net/downloads/wwf_marine_briefing_principles_blue_economy.pdf on 1-11-2019.
Food and Agriculture Organization	2018	Blue Economy encompasses sustainable development of aquatic-based industries, including fisheries (capture and culture), fish processing, transport, tourism, energy, etc. Similar to the concept of green economy, the blue economy looks into maximizing the socioeconomic benefits of aquatic industries while minimizing environmental degradation	FAO (2018). FAO joins governments in affirming commitment to Blue Economy. Retrieved from http://www.fao.org/africa/news/detail-news/en/c/1172093/ on 28-10-2019
Conservation International	2018	At its simplest, "Blue Economy" refers to the range of economic uses of ocean and coastal resources — such as energy, shipping, fisheries, aquaculture, mining, and tourism. It also includes economic benefits that may not be marketed, such as carbon storage, coastal protection, cultural values and biodiversity.	Bertazzo, S. (2018). What on Earth is the 'blue economy'? Retrieved from https://www.conservation.org/blog/what-on-earth-is-the-blue-economy/?gclid=Cj0KCQjwl9zdBRDgARIsAL5Nyn3xGXHsApcgFjjO6CvN0Zg602NYYuJw2LPvqa_nDpKKxPtNDJWQxLYaAvSLEALw_wcB on 21-10-2019

Keen et al.	2018	The Blue Economy aims to balance sustainable economic benefits with long-term ocean health, in a manner which is consistent with sustainable development and its commitment to intra- and intergenerational equity.	Keen, R., Schwarz, A. & Wini-Simeon, L. (2018). Towards defining the Blue Economy: Practical lessons from Pacific Ocean governance. Marine Policy, 88, 333-341.
United Nations Development Programme	2018	The Blue Economy is an approach put forward by the international community to take into account the health of the oceans and seas as we strive to balance the three dimensions of sustainable development: economic, social and environmental. This concept promotes economic growth, social inclusion and improved livelihoods at the same time as ensuring the environmental sustainability of oceans and seas.	UNDP (2018). Blue Economy: Community Solutions. UNDP, New York.
European Commission	2019	For the purpose of this report, the EU's Blue Economy encompasses all sectoral and cross-sectoral economic activities related to the oceans, seas and coasts, including those in the EU's outermost regions and landlocked countries. [..] It comprises emerging sectors and economic value based on natural capital and non-market goods and services.	European Commission (2019). The EU Blue Economy Report 2019. Publications Office of the European Union. Luxembourg.
United Nations Conference on Trade and Development	2019	The Oceans economy encompasses a sustainable economy for the ocean-based marine environment, related biodiversity, ecosystems, species and genetic resources including marine living organisms (from fish and algae to micro-organisms) and natural resources in the seabed, while ensuring their sustainable use and hence, conservation.	UNCTAD (2019). Oceans Economy and Fisheries. Retrieved from https://unctad.org/en/Pages/DITC/Trade-and-Environment/Oceans-Economy.aspx on 22-10-2019
World Bank	2019	The term Blue Economy refers to the development of oceanic economic activities in an integrated and sustainable way. It is focused on capturing potential synergies and managing the trade-offs across industries to better address the growing threats now confronting oceans, and particularly those posed by climate change. A Blue Economy strives to maximize the socio-economic benefits generated by oceanic activities, including food production and job creation.	World Bank. (2019). The World Bank's Blue Economy Program and PROBLUE: Supporting integrated and sustainable economic development in healthy oceans. Retrieved from https://www.worldbank.org/en/topic/environment/brief/the-world-banks-blue-economy-program-and-problue-frequently-asked-questions on 24-10-2019
European Commission	2019	Blue Growth is the long-term strategy to support sustainable growth in the marine and maritime sectors as a whole.	European Commission (2019). Blue growth. Retrieved from https://ec.europa.eu/maritimeaffairs/policy/blue_growth_en on 24-10-2019

Annex 2: Global blue economy sectors across the literature

The sectors are based on the established and emerging sectors identified in the 2018 annual economic report on the EU blue economy[1]. Sectors highlighted in blue are emerging sectors identified by the research team.

Table 2: Global blue economy sectors

		EUROPEAN UNION			UNITED NATIONS		OTHERS	
Sectors	**Subsectors**	**European Commission 2018[2]**	**European Commission 2019[3]**	**European Environment Agency (EEA)[4]**	**UNEP/Nairobi Convention, 2015[5]**	**UNECA, 2016[6]**	**OECD 2016[7]**	**World Bank**
Non-Living Resources	**Extraction of minerals**	Deep-sea mining	Extraction of marine aggregates / Marine minerals	Extraction of minerals	Coastal mining; Dredging	Deep sea and seabed mining - Coastal mining of sand, gravel, and other construction materials	Marine and seabed mining	Minerals, seabed mining
	Extraction of salt			Extraction of salt	The production of sea salt from saltpans typically located on estuary flood plains	Extraction of salt	Seawater dissolved minerals extraction	
	Extraction of oil and gas	Marine extraction of oil & gas, including support activities and gas	Marine extraction of oil & gas, including support activities and gas	Extraction of oil and gas	Fossil fuels from the coastal zone and offshore	Extraction of oil and gas	Offshore oil and gas	Oil and gas
	Offshore structures (incl. submarine cables, pipelines operations and decommissioning of platforms)	Multi-use platforms	Multi-use platforms	Offshore structures			Marine manufacturing and construction	

[1] The 2018 annual economic report on the EU blue economy: https://op.europa.eu/en/publication-detail/-/publication/79299d10-8a35-11e8-ac6a-01aa75ed71a1
[2] https://op.europa.eu/en/publication-detail/-/publication/79299d10-8a35-11e8-ac6a-01aa75ed71a1
[3] https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en
[4] European Environment Agency (EEA), "Pressures and their effects in EU seas" (in prep)
[5] http://wedocs.unep.org/bitstream/handle/20.500.11822/9668/-Regional_State_of_the_Coast_Report_Western_Indian_OceanRSOCR_Final.pdf.pdf?sequence=2&isAllowed=y
[6] https://www.uneca.org/publications/africas-blue-economy-policy-handbook
[7] https://geoblueplanet.org/wp-content/uploads/2016/05/OECD-ocean-economy.pdf

	Extraction of water - Desalination	Desalination	Desalination	Extraction of water			Desalination	Desalination
	Storage of CO2 / Carbon sequestration	Carbon sequestration	Carbon sequestration				Carbon capture storage	Carbon Sequestration, blue carbon
	Marine litter removal	Waste recycling and storing	Waste recycling and storing					
	Storage of gas (EEA)							
Living Resources	**Fish and shellfish harvesting**	Fisheries	Fisheries	Fish and shellfish harvesting	Artisanal and industrial fishing	Fish and shellfish harvesting	Capture fisheries	Fisheries
	Fish and shellfish processing	Processing and distribution	Processing and distribution	Fish and shellfish processing	Fish processing	Sporadically mentioned, not considered a key principle (table 1; pg.6)	Seafood processing	
	Marine plant and algae harvesting		Algae sector	Marine plant and algae harvesting		Marine plants		
	Hunting and collecting for other purposes			Hunting and collecting for other purposes				
	Aquaculture	Aquaculture	Aquaculture	Aquaculture	Aquaculture	Aquaculture	Marine aquaculture	Aquaculture
	Blue bioeconomy	Blue economy / Biotechnology	Blue bio economy	Blue biotechnology	Marine Genetic Resources and Bioprospecting	Pharmaceuticals, chemicals, cosmetics, genetic research	Marine biotechnology	Marine biotechnology - Pharmaceuticals, chemicals
Renewable Energy	**Renewable energy generation**	Offshore wind energy / ocean wave-tidal energy	Offshore wind / ocean energy (tidal and wave power)	Renewable energy generation	Wind		Offshore wind energy / Ocean renewable energy	Energy - Renewables
	Transmission of electricity and communications			Transmission of electricity and communications				
Maritime Transport	**Transport infrastructure**	Ports, warehousing and water projects	Port activities / Warehousing and storage / Cargo handling / Construction of water projects / Service activities incidental to water transportation	Transport infrastructure	Ports (and activities taking place here) are a major aspect that are discussed.	Transport infrastructure	Ports	Transport and trade - Port infrastructure and services

	Restructuring of seabed morphology			Restructuring of seabed morphology				
	Transport shipping	Maritime transport / Shipping / Sea and coastal transport	Maritime transport	Transport - shipping	Shipping	Transport - shipping	Shipping	Transport and trade - Shipping
	Shipbuilding and repair	Shipbuilding and repair	Shipbuilding and repair			Shipbuilding and repair	Shipbuilding and repair	
Coastal Tourism and Leisure	**Tourism and leisure infrastructure**	Coastal tourism	Coastal tourism	Tourism and leisure infrastructure	Tourism and ecotourism	Tourism and leisure infrastructure	Maritime and coastal tourism	Tourism and recreation
	Tourism and leisure activities	Coastal tourism	Coastal tourism	Tourism and leisure activities	Tourism and ecotourism	Tourism and leisure activities	Maritime and coastal tourism	Tourism and recreation
Public Sector	**Military operations**	National defence, coast guards	Maritime defence	Military operations				
	Research, survey and educational activities	Marine education and research	Marine education and research	Research, survey and educational activities	Coastal and Marine Research and Capacity Building	Knowledge and information	Marine R&D and education	Ocean monitoring and surveillance - Technology and R&D
	Land claim			Land claim				
	Canalisation and other watercourse modifications			Canalisation and other watercourse modifications				
	Coastal defence and flood protection	Coastal and environmental protection / coastal and habitat protection / defence	Coastal protection	Coastal defence and flood protection	Marshes and mangroves	Coastal protection		Coastal Protection
	Waste removal	Waste recycling and storing	Waste recycling and storing	Waste treatment and disposal	Waste management			Waste Disposal

Annex 3: Threats and trends affecting blue economy sectors

This annex describes the most important threats and trends identified for each blue economy sector and subsector, as discussed in Chapter 2.3.

Table 3 describes the main threats and risks specific to each blue economy sector and subsector. The table also includes specific examples of local or global occurrences of the main threats or risks facing each subsector. To the extent possible, the threats and risks for each subsector are displayed in order of significance. Geographic dimension is provided where possible, based on literature review. While research has been conducted to ensure that the most significant risks and threats and the most illustrative examples are covered, the table should not be considered exhaustive. The table feeds into Tasks 5 and 6.

Table 3: Main threats of the blue economy sectors

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
Non-Living Resources	**Extraction of minerals**	Limited knowledge on environmental impacts: ecosystem health, species abundance, biodiversity and habitat impacts (i.e. loss) through seafloor disturbance, sediment plumes, vibration, pollution from noise, light, leaks/spills of fuel, toxic products, and other surface accidents.[8,9,10,11,12,13]	X	X				X	X	X		X	Atlantis II project in the Red Sea: considered one of the most promising sites for deep-sea mining. It is, however, held up by legal and political disputes. [14] Bismarck Sea: The Solwara 1 project was expected to become the world's first deep-sea mining project. In September 2019,

[8] https://www.nature.com/articles/ngeo2983; https://www.springer.com/gp/book/9783030126957
[9] https://portals.iucn.org/library/sites/library/files/documents/2018-029-En.pdf; https://www.springer.com/gp/book/9783030126957
[10] https://www.nioz.nl/en/valorisation/deep-sea-science-and-technology; https://link.springer.com/chapter/10.1007/978-3-319-60156-4_11
[11] https://www.oceanfdn.org/sites/default/files/Independent%20Review%20of%20the%20Environmental%20Impact%20Statement%20for%20the%20proposed%20Nautilus%20Mineral%20Solwara%201%20Seabed%20Mining%20Project%20Papua%20New%20Guinea.pdf
[12] https://www.iucn.org/resources/issues-briefs/deep-sea-mining; https://www.frontiersin.org/articles/10.3389/fmars.2017.00418/full#B76
[13] https://link.springer.com/chapter/10.1007/978-3-319-60156-4_11; https://portals.iucn.org/library/sites/library/files/documents/2018-029-En.pdf
[14] https://portals.iucn.org/library/sites/library/files/documents/2018-029-En.pdf

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		Alteration of carbon cycling, carbon transport and sediment sinks (important deep-sea ecosystem services)[16,17].	X	X				X	X	X		X	the project nevertheless collapsed due to community resistance, legal challenges and funding difficulties. [15,]
		Uncertainties at each stage of the value chain, coupled with fluctuating metal prices, challenge the profitability of deep-sea mining.[18]	X					X	X	X		X	
		Circular economy (reduced demand for raw materials).[19]	X	X			X						
		Costly and advanced technology and mapping methods requirements.[20,21]	X							X			
		Challenges for establishing a comprehensive and dynamic regulatory framework that protects the interests of several stakeholders, especially small-scale fisheries and tourism. [22,23]	X		X	X	X	X	X	X		X	
		Challenging new locations (e.g. the Arctic and deep-sea> 3000). [24]	X	X				X	X	X		X	Accessing Artic presents high costs and technological challenges. The extraction of for polymetallic nodules deposits located at greater depths in the Pacific Ocean (>3000m) poses great technological challenges.[25,26]

[16] https://www.ncbi.nlm.nih.gov/pubmed?Db=pubmed&Cmd=ShowDetailView&TermToSearch=21688032
[17] https://www.dosi-project.org/wp-content/uploads/2019/07/040-DOSI-Climate-change-considerations-V4.pdf
[15] https://digital-library.theiet.org/content/journals/10.1049/et.2019.0202; https://www.theguardian.com/world/2019/sep/16/collapse-of-png-deep-sea-mining-venture-sparks-calls-for-moratorium
[18] https://pubs.geoscienceworld.org/msa/elements/article/14/5/301/559105/Deep-Ocean-Mineral-Deposits-Metal-Resources-and; https://portals.iucn.org/library/sites/library/files/documents/2018-029-En.pdf
[19] https://portals.iucn.org/library/sites/library/files/documents/2018-029-En.pdf
[20] https://www.msp-platform.eu/sector-information/marine-aggregates-and-marine-mining; https://www.ingentaconnect.com/content/mts/mtsj/2011/00000045/00000005/art00004;jsessionid=dp0pmejqgm5cc.x-ic-live-02
[21] https://pubs.geoscienceworld.org/msa/elements/article/14/5/301/559105/Deep-Ocean-Mineral-Deposits-Metal-Resources-and
[22] https://pubs.geoscienceworld.org/msa/elements/article-abstract/14/5/331/559116/Deep-Sea-Mining-International-Regulatory?redirectedFrom=fulltext
[23] https://pubs.geoscienceworld.org/msa/elements/article/14/5/301/559105/Deep-Ocean-Mineral-Deposits-Metal-Resources-and
[24] https://www.researchgate.net/publication/322686290_CHALLENGES_OF_DREDGING_IN_THE_ARCTIC_AND_OTHER_DEEP_OCEAN_LOCATIONS
[25] https://webgate.ec.europa.eu/maritimeforum/en/file/8090/download?token=n9TOblr-
[26] https://www.frontiersin.org/articles/10.3389/fmars.2017.00418/full

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		Competition over access to resources and space with other maritime activities.[27,28]	X	X		X		X	X			X	
		Emergence of environmental protection and conservation regimes and organisations Creation of bodies of indigenous and human rights law.[29,30]	X	X	X	X	X						An NGO working with coastal communities near the site of planned undersea mining in Papua New Guinea's north is demanding a Pacific-wide ban on seabed mining.[31]
	Extraction of salt	See "Extraction of minerals"; many articles qualify ''extraction of salt'' as similar to ''extraction of minerals'', i.e. what is mentioned above also accounts to this subsector											
		Negative effect on growth and stocking of mangrove forests[32]		X				X	X			X	A study concluded that salt production poses threats to the sustainability of mangrove forestry ecosystems on the south western Indian Ocean cost in Tanzania[33]
		Microplastics found in salt products for human consumption could affect human health, with potential economic impacts. [34]	X	X	X		X						A research study showed that microplastics concentration in sea salts was significantly higher in lake salts and rock / well salts. This finding suggests that microplastics contaminate marine goods, such as sea salts.[35]

[27] https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
[28] https://webgate.ec.europa.eu/maritimeforum/en/file/8090/download?token=n9TOblr-
[29] https://harvardelr.com/2018/04/16/broadening-common-heritage/#_ftnref43
[30] https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
[31] https://www.rnz.co.nz/international/pacific-news/404425/ngo-calls-for-pacific-wide-ban-on-seabed-mining
[32] https://pdfs.semanticscholar.org/7e88/9a4601bfb8bcbeb8d632f41811caa474820c.pdf
[33] https://pdfs.semanticscholar.org/7e88/9a4601bfb8bcbeb8d632f41811caa474820c.pdf
[34] salt/https://pdfs.semanticscholar.org/7e88/9a4601bfb8bcbeb8d632f41811caa474820c.pdf
[35] https://pubs.acs.org/doi/abs/10.1021/acs.est.5b03163

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
	Extraction of oil and gas	Fluctuations in oil prices. Oil and gas prices are crucial factors in deciding whether a reserve is economically feasible.[36]	X				X						A research study found economic impacts of oil price fluctuations in developed and emerging economies such as China. [37]
		Handling governments' pressure related to carbon reduction targets[38].	X	X		X	X						The EU's 2050 energy strategy aims to reduce greenhouse gas emissions by 2050 having a global effect.[39]
		Expansion of oil & gas industry towards deeper waters (>200 m) and new regions (e.g. Artic) in the absence of sufficient baseline data and knowledge, will affect marine ecosystems and increase of costs.[40,41]	X	X				X	X	X		X	Accessing Artic presents serious technological challenges (i.e. deep sea, offshore drilling in extreme weather conditions); high risk of oil spills and challenging spill recovery; complicated transportation.[42]
		Trade wars and protectionism may lead to slower economic growth and lower O&G prices. [43]	X		X	X	X						
		Increased focus on natural gas instead of oil. [44]	X				X						
		Natural gas hydrates extraction poses technological (complex and costly), environmental (affect seafloor stability and release methane) and governance (no	X	X			X						

[36] https://www.aeaweb.org/articles?id=10.1257/jep.30.1.139
[37] https://pdfs.semanticscholar.org/ec5b/447396d15390aaae4b43eb7a95a6de576eaf.pdf
[38] https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
[39] https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
[40] https://www.frontiersin.org/articles/10.3389/fenvs.2016.00058/full; https://www.msp-platform.eu/sector-information/oil-and-gas; https://www.thermofisher.com/blog/mining/arctic-oil-gas-exploration-do-the-challenges-outweigh-the-potential/
[41] https://worldoceanreview.com/en/wor-3/oil-and-gas/; http://www.iploca.com/platform/content/element/30121/GlobalUpstreamTrendssubmittedFinal.pdf
[42] https://www.thermofisher.com/blog/mining/arctic-oil-gas-exploration-do-the-challenges-outweigh-the-potential/
[43] http://www.iploca.com/platform/content/element/30121/GlobalUpstreamTrendssubmittedFinal.pdf
[44] https://oilprice.com/Energy/Energy-General/The-Next-Big-Trend-In-Offshore-Oil-Gas.html

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			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		coordinated deep-sea international regulations) challenges.[45,46]											
		Competition over access to resources and space with other maritime activities including aquaculture, fishing, shipping, offshore wind and extraction of minerals[47,48]	X	X		X	X						
		Social acceptance[49,50]	X		X		X						Low acceptance with NGOs regarding O&G in the Arctic.[51]
		Decommissioning (see "offshore infrastructures")[52]	X	X	X			X	X			X	
	Offshore structures (incl. submarine cables, pipelines operations and decommissioning of platforms)	Offshore constructions are prone to high technical risks. Natural events can cause hazardous materials releases, endangering workers, the population and the environment.[53]	X	X	X			X	X			X	In Europe, most incidents at fixed offshore structures have involved falling loads for both metal jacket and concrete base platforms due to storms, while in areas such as the Gulf of Mexico tropical storms are likely to lead to severe direct damage, resulting in platform capsizing, sinking or grounding.[54]
		Understanding of the environmental effects of different decommissioning strategies is still incomplete.[55]	X	X				X	X			X	From 2017 to 2025, decommissioning is set to occur on 349 fields across the North Sea. Removal is desirable as old cables and pipelines can impede other uses of

[45] https://res.mdpi.com/d_attachment/geosciences/geosciences-09-00443/article_deploy/geosciences-09-00443.pdf; https://agupubs.onlinelibrary.wiley.com/doi/full/10.1002/2016RG000534
[46] https://www.primeast.com/top-5-challenges-facing-oil-and-gas-companies/;
[47] https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/; https://www.frontiersin.org/articles/10.3389/fmars.2017.00418/full
[48] https://www.frontiersin.org/articles/10.3389/fmars.2017.00418/full
[49] https://www.eldor.no/blog/will-the-oil-and-gas-industry-be-sustainable-in-the-future; https://oilprice.com/Energy/Energy-General/The-Next-Big-Trend-In-Offshore-Oil-Gas.html;
[50] https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
[51] https://thinkprogress.org/anwr-advocates-write-letters-oil-gas-drilling-arctic-f5d22e55afaf/
[52] https://www.msp-platform.eu/sector-information/oil-and-gas
[53] https://ui.adsabs.harvard.edu/abs/2017EGUGA..1917733N/abstract; https://link.springer.com/chapter/10.1007/978-3-319-51159-7_5
[54] https://ui.adsabs.harvard.edu/abs/2017EGUGA..1917733N/abstract
[55] https://www.sciencedirect.com/science/article/pii/S0048969718350551

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			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		Cable and pipes spatial planning conflicts especially with extractive uses (i.e. marine aggregates, oil and gas extraction, fishing, etc.)[58]	X	X		X	X	X	X			X	the seabed, yet its environmental impacts remain unknown.[56 57]
	Extraction of water - Desalination	Given the energy-intensive processes involved in desalination, the global emissions reduction targets pose a serious risk. More desalination plants will be necessary as the planet warms, increasing in turn GHG emissions.[59,60,61]	X	X				X	X			X	Saudi Arabia produces the largest amount of desalinated water in the world, due to limited fresh water supply and cheap energy costs for fossil fuels. In the Middle East, desalination plants produce a particularly high amount of GHG emissions, as fossil fuels are the main source for electricity generation.[62]
		Waste disposal of by-products may impact animal life (e.g. brine production).[63]		X				X	X			X	A research study revealed brine production to be approximately 50% greater than previous quantifications, causing a more significant environmental impact than expected. [64]
		Organism mortality caused by seawater processing.[65]		X				X	X			X	Carlsbad plant, California, USA: The plant draws in small organisms amounting to 1.5 pounds of adult fish per day, causing a significant environmental impact. The plant also has high economic costs (1 billion to build, with a rough estimate of
		High costs of constructing plants and processing water[68]	X					X				X	

[58] https://www.msp-platform.eu/sites/default/files/sector/pdf/mspforbluegrowth_sectorfiche_cablespipelines.pdf
[56] https://www.msp-platform.eu/sites/default/files/sector/pdf/mspforbluegrowth_sectorfiche_cablespipelines.pdf
[57] https://www.boem.gov/sites/default/files/oil-and-gas-energy-program/Leasing/Regional-Leasing/Pacific-Region/Fowler_et_al-2018-Frontiers_in_Ecology_and_the_Environment-%281%29.pdf
[59] https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
[60] https://e360.yale.edu/features/as-water-scarcity-increases-desalination-plants-are-on-the-rise
[61] https://www.sciencedirect.com/science/article/pii/S0048969718349167
[62] https://e360.yale.edu/features/as-water-scarcity-increases-desalination-plants-are-on-the-rise
[63] http://news.mit.edu/2019/brine-desalianation-waste-sodium-hydroxide-0213; https://imem.ua.es/va/documentos/arxius-imem/articles-investigadors/jose-luis-sanchez-lizaso/belatoui-et-al-2017.pdf;
[64] https://www.sciencedirect.com/science/article/pii/S0048969718349167?via%3Dihub#!
[65] https://e360.yale.edu/features/as-water-scarcity-increases-desalination-plants-are-on-the-rise
[68] https://e360.yale.edu/features/as-water-scarcity-increases-desalination-plants-are-on-the-rise

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			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
													$50 million a year for the power to run it).[66,67]
	Storage of CO_2 / Carbon sequestration	Social acceptance[69]			X								In the North Sea, the first demonstration project for CCS was introduced in 2004 (K12-B)[70]
		Economic feasibility (upscaling and reducing costs, inadequate investment)[71,]	X				X						
		Technological feasibility[72]	X				X						
		Obtaining geological information from areas explored or used by oil and gas companies.[73]	X				X						
		Lack of legal and regulatory frameworks[74]				X							
	Marine litter removal	Challenges related to the governance of a global, cross-sectoral approach[75]				X	X						The environmental impacts of large systems, such as the one of Ocean Cleanup, are not researched in detail.[76]
		End of pipe solution criticised by some parties[77]			X	X	X						
		Difficulties of removing waste that sinks to the seabed. Although data is still quite limited, it has been suggested that quantities on the seabed may exceed 10.000 items per hectare[78]		X			X						
	Storage of gas	Not specified for marine environments, only the sector as a whole where key challenges	X				X						Regions of seismic activity.[80]

[66] https://e360.yale.edu/features/as-water-scarcity-increases-desalination-plants-are-on-the-rise
[67] https://www.kqed.org/science/1115545/desalination-why-tapping-sea-water-has-slowed-to-a-trickle-in-california
[69] https://www.sciencedirect.com/science/article/pii/S0959652614009536
[70] https://sequestration.mit.edu/tools/projects/k12b.html
[71] https://www.intechopen.com/books/recent-advances-in-carbon-capture-and-storage/economics-of-carbon-capture-and-storage; https://ec.europa.eu/info/sites/info/files/iogp_-_report_-_ccs_ccu.pdf
[72] https://www.frontiersin.org/articles/10.3389/fenrg.2014.00055/full
[73] https://ec.europa.eu/clima/sites/clima/files/strategies/progress/docs/com_2015_576_annex_2_en.pdf
[74] https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
[75] https://eos.org/meeting-reports/a-cross-sectoral-approach-to-tackle-ocean-plastic-pollution; https://onlinelibrary.wiley.com/doi/abs/10.1111/rec.12388; https://timera-energy.com/the-challenges-of-investing-in-gas-storage/
[76] https://www.wur.nl/en/newsarticle/Frequently-asked-questions-about-The-Ocean-Cleanup.htm
[77] https://www.reuters.com/article/us-g20-summit-plastics/g20-plastic-trash-reduction-goal-doesnt-address-excessive-production-activists-idUSKCN1TW1OD
[78] https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
[80] https://geocenter.info/en/monitoring/earthquake

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			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		relate to the seasonality and economic viability[79,]											
Living Resources	**Fish and shellfish harvesting**	Degradation of aquatic ecosystems, alterations of biodiversity and food web structures caused by overfishing, pollution, and destructive fishing (e.g. bycatch)[81,82,83.]	X	X	X		X						The Northern Adriatic Sea, the Black Sea, the Baltic Sea, the northern Gulf of Mexico and the Chesapeake Bay are seriously affected by coastal hypoxia. In the northern Gulf of Mexico, the seasonal hypoxic zone poses serious threats to the gulf's seafood industry. In the Mediterranean Sea, 85 % of the discharged sewage is untreated, which leads to eutrophication and the creation of dead zones. [84] In the North Sea, fishermen compete over space with new offshore wind farms[85]
		Effects of climate change on the global marine ecosystems.[86] > Global warming on oxygen levels in the oceans and the effect on ocean nutrient cycles and marine habitat.[87,]	X	X	X		X						
		Stock depletion (overfishing) stock fluctuations differences in profitability between basins; low average sale prices for many commercially important species.[88,89,90,91]	X	X			X						
		Effects of agricultural runoff of nutrient and phosphorus leading to eutrophication and dramatic rise in coastal hypoxia, which poses a great threat to coastal ecosystems.[92,]	X	X	X		X						

[79] https://www.businesswire.com/news/home/20190919005323/en/Global-Natural-Gas-Storage-Market-Outlook-2026 ; https://www.sciencedirect.com/science/article/pii/S235285401730089X
[81] https://www.seafoodsource.com/news/environment-sustainability/7-challenges-facing-future-global-seafood-supply
[82] https://www.msp-platform.eu/sites/default/files/sector/pdf/mspforbluegrowth_sectorfiche_fishing.pdf
[83] https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en; https://www.seafoodsource.com/news/environment-sustainability/7-challenges-facing-future-global-seafood-supply
84 https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
85 https://www.politico.eu/article/fishermen-offshore-wind-farms-struggle-to-share-sea/
86 https://www.seafoodsource.com/news/environment-sustainability/7-challenges-facing-future-global-seafood-supply
87 https://www.nature.com/articles/nature21399; https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
88 https://our.fish/press/eu-despite-reformed-common-fisheries-policy-were-still-overfishing/; https://stecf.jrc.ec.europa.eu/documents/43805/2483556/STECF%2019-06%20-%20AER%20-%202019.pdf
89 https://helcom.fi/action-areas/fisheries/commercial-fisheries/; https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
90 https://stecf.jrc.ec.europa.eu/documents/43805/2483556/STECF%2019-06%20-%20AER%20-%202019.pdf
91 https://stecf.jrc.ec.europa.eu/documents/43805/2192243/STECF+18-19+-+EU+Aquaculture+Economics.pdf/dc9c871e-830e-477f-aec8-5252ac102e19
92 https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en

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			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		Poor fisheries management (regulations and transparencies).[93]	X	X	X	X	X						Regional policies can fail to improve fisheries sustainability. For example, after the European Union reform of the Common Fisheries Policy, overfishing is still present in European waters, less than 20% of stocks are exploited sustainably in the Mediterranean.[94.]
		Impact and competition over resources access and space of other maritime activities.[95,96,97]	X	X	X	X	X						
		Challenges related to the fishing governance of IUU, discards and bycatch.[98]	X	X	X	X	X						IUU fishing occurs not only in the high seas but also within exclusive economic zones (EEZs) that are poorly managed.[99]
		Risks of social disruption due to changes in fishery regulations, fishing industry technology, resource abundance.[100]	X		X		X						The decline of fishing stocks causes social impacts at a regional level, as for the cod fishery disruption in the US, or at a local level, where overfishing can destroy the livelihood of local communities, particularly in third world countries[101,102].
		Impact of industrial fleets in small-scale fisheries, which produce two-thirds of catches destined for direct human consumption and provide 90% of the employment in the sector.	X		X	X	X						

93 http://www.fao.org/3/a-i0584e.pdf
94 https://www.sciencedirect.com/science/article/pii/S0308597X17307364; https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
95 https://www.msp-platform.eu/sites/default/files/sector/pdf/mspforbluegrowth_sectorfiche_fishing.pdf
96 https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
97 https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
98 https://fas.org/irp/nic/fishing.pdf; http://www.fao.org/3/CA2905EN/ca2905en.pdf; http://www.fao.org/3/CA2905EN/ca2905en.pdf
99 http://www.fao.org/3/Y3274E/y3274e08.htm
100 http://www.fao.org/3/W1238E05.htm
101 https://www.pnas.org/content/116/46/22912
102 https://www.un.org/africarenewal/magazine/may-july-2017/overfishing-destroying-livelihoods

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			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		Small-scale fisheries are considered central to the success and sustainability of the blue economy.[103,104]											
		Association of the seafood industry with labour and human rights issues causing economic losses.[105]	X		X	X	X	X	X			X	Rights abuses and forced labour were reported in Thailand's Fishing Industry, the EU stopped imports coming from there.[106,107]
	Fish and shellfish processing	See "Fish and shellfish harvesting"; sectors are so interrelated that what accounts for above, subsequently impacts this sector	X	X	X	X	X						
	Marine plant and algae harvesting	Strong challenges remain in algae production (e.g. strain isolation, nutrient sourcing, harvesting). This needs to be addressed first before it can compete with other resources (e.g. petroleum)[108,109]	X				X						
		Seaweed farming risks: facilitation of disease, alteration of population genetics and wider alterations to the local physiochemical environment, contamination with pesticides/algicides. Acidification[110]	X	X				X	X				In the Philippines (2011 & 2013) a bacteria outbreak affecting a valuable seaweed species caused a devastating loss to the communities involved, estimated at over US$ 310 million.[111]
		Communities depending on seaweed farming become highly vulnerable to disease	X	X	X	X		X	X			X	

103 http://www.fao.org/policy-support/policy-themes/sustainable-small-scale-fisheries/en/; https://www.nature.com/articles/ncomms10244
104 https://www.iied.org/focusing-blue-economy-future-small-scale-fisheries
105 https://www.hrw.org/report/2018/01/23/hidden-chains/rights-abuses-and-forced-labor-thailands-fishing-industry
106 https://www.un.org/africarenewal/magazine/may-july-2017/overfishing-destroying-livelihoods
107 https://www.hrw.org/report/2018/01/23/hidden-chains/rights-abuses-and-forced-labor-thailands-fishing-industry
108 https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5836383/
109 https://www.intechopen.com/books/algae-organisms-for-imminent-biotechnology/challenges-and-opportunities-in-the-present-era-of-marine-algal-applications
110 https://www.frontiersin.org/articles/10.3389/fmars.2019.00107/full; https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
111 https://onlinelibrary.wiley.com/doi/full/10.1111/jwas.12649

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			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		outbreaks or meteorological natural disasters. [112]											
		Competition over access to maritime space with other maritime activities. [113]	X	X	X			X	X			X	
	Hunting and collecting for other purposes												
	Aquaculture	Short-term threats: disease, antibiotic use, algal blooms, pollution[114 115]; Medium-term threats: greenhouse gas emissions. ecosystem degradation, habitat destruction (i.e. mangrove forests, biodiversity loss and genetic contamination (escaped fish).[116,117,118,119]	X	X	X			X	X			X	In northern parts of Europe, warmer water and salinity changes will likely result in increased salmon diseases and parasites, as well as more algal blooms.[120] In tropical areas, mangrove forests have been cut down and replaced with ponds for shrimp farming. High costs and decreasing returns resulting from erosion, pollution and disease leads to many of these ponds to be abandoned after a few years, and new ponds to be built elsewhere. Apart from the income insecurity to the farmers, mangrove forests act as coastal protection, fish nursery grounds, filters and carbon sinks,
		Climate change: Natural disasters (e.g. extreme weather and floods) and	X	X			X						

112 https://onlinelibrary.wiley.com/doi/full/10.1111/jwas.12649
113 https://blogs.unb.ca/newsroom/_media/images/2016/09/2482893c83ca06153f5dc3a627a59d6bcottier-cook-briefing-1.pdf
114 https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0042724
115 https://ec.europa.eu/environment/integration/research/newsalert/pdf/sustainable_aquaculture_FB11_en.pdf
116 https://www.pyxeraglobal.org/snv-integrates-shrimp-aquaculture-mangrove-protection-ca-mau-vietnam/
117 https://www.fairr.org/article/shallow-returns-esg-issues-in-aquaculture/
118 https://www.nature.com/articles/s41558-019-0425-9
119 https://academic.oup.com/icesjms/article/73/10/2488/2647116
120 http://www.mccip.org.uk/media/2031/21_aquaculture_2020.pdf

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			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		temperature increase, acidification, sea-level rise.[122,123,124,125]											and their destruction can have significant detrimental impact.[121]
		Competition over access to resources and space with other maritime activities. Inappropriate spatial site selection (Near- and offshore) threats the sustainable development and expansion of the industry (Marine spatial planning).[126]	X	X	X	X	X						Potential Risks and Synergies between (bivalve) aquaculture and other marine offshore activities exist in the Dutch North Sea, in terms of technical, economic, ecological and governance aspects[127].
		Transparency and food fraud[128]	X		X	X	X						A class-action lawsuit filed on behalf of U.S. direct purchasers of Norwegian farmed salmon accused multiple Norwegian firms of conspiring to fix the prices of farmed salmon.[129]
		Community resistance. Marine aquaculture requires social acceptance.[130,131]			X	X	X						In December 2018, the government of British Columbia reached an agreement with First Nations peoples to close several Mowi and Cermaq farms and restrict salmon farming in coastal areas, which play an important role in their livelihood.[132]

122 http://www.fao.org/3/i2727e/i2727e02.pdf
123 https://onlinelibrary.wiley.com/doi/abs/10.1111/raq.12140
124 https://link.springer.com/article/10.1007/s00267-018-1117-3
125 https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
121 https://www.seafoodwatch.org/ocean-issues/aquaculture/habitat-damage;
126 https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
127 https://link.springer.com/article/10.1007/s10499-016-9987-y
128 https://usa.oceana.org/reports/oceana-study-reveals-seafood-fraud-nationwide
129 https://www.seafoodsource.com/news/business-finance/us-companies-file-class-action-suit-alleging-price-fixing-by-norwegian-farmed-salmon-firms
130 https://www.cbc.ca/news/canada/british-columbia/bc-salmon-farming-lawsuit-1.4976042
131 https://www.tandfonline.com/doi/abs/10.1080/23308249.2015.1121202?src=recsys&journalCode=brfs21
132 https://www.cbc.ca/news/canada/british-columbia/bc-salmon-farming-lawsuit-1.4976042

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
	Blue bioeconomy	Space and ecosystem pressure[133]	X	X	X	X	X						
		Presence of heavy metals in the water[134]		X			X						
		Financing is a major issue for blue bioeconomy projects, start-ups and SMEs[135]	X				X	X					
		Potential loss of jobs, particularly in developing countries (need for high-skilled jobs). [136]	X		X		X						
		High costs of exploration (technological) and production [137].	X				X						
Renewable Energy	**Renewable energy generation / Transmission of electricity and communications**	Wind energy: Noise, electromagnetic interference, animal safety, visual effects[138]		X			X						Wind turbines have caused increased mortality rates in migratory birds and bats, particularly in migratory bottlenecks and near breeding colonies[139] In the North sea, the installation (pile-driving) of foundations may involve underwater noise, which impacts the harbour porpoises[140]
		Solar energy: Hazardous materials. Increases in temperature will reduce output efficiency[141,142]	X	X			X						
		Hydropower: Disruption of river ecology, damaging of forests and biodiversity, release of GHGs, disruption of food systems and agriculture, deterioration of water quality. Changes in precipitation.[143,144]		X			X						

133 https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2018EF001014
134 https://webgate.ec.europa.eu/maritimeforum/en/file/10873/download?token=8thGZmGO
135 https://op.europa.eu/en/publication-detail/-/publication/79299d10-8a35-11e8-ac6a-01aa75ed71a1; https://webgate.ec.europa.eu/maritimeforum/en/file/11296/download?token=em6HzmR0
136 https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2018EF001014
137 https://webgate.ec.europa.eu/maritimeforum/en/file/11296/download?token=em6HzmR0
138 https://www.researchgate.net/publication/281550386_Environmental_Impacts_of_Renewable_Energy_Technologies
139 https://royalsocietypublishing.org/doi/10.1098/rspb.2017.0829; https://www.boem.gov/sites/default/files/renewable-energy-program/State-Activities/RiskBirdsWildlifeOffshore.pdf
140 http://publications.naturalengland.org.uk/publication/4813967957950464
141 https://www.researchgate.net/publication/281550386_Environmental_Impacts_of_Renewable_Energy_Technologies
142 https://www.sciencedirect.com/science/article/pii/S1364032119306239
143 https://www.governmenteuropa.eu/environmental-impact-of-hydropower/91552/
144 https://www.sciencedirect.com/science/article/pii/S1364032119306239

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		Potential competition or conflicts with other maritime uses[145].	X	X	X	X							In Belgium, all non-maintenance vessels have to remain at least 500 meters away from wind parks. According to fishermen these restrictions limit their ability to fish sustainably.[146]
Maritime Transport	**Transport infrastructure**	Local and global air pollution; Noise; Vibration; Water pollution; Water turbidity; Soil/sediment Pollution; GHG and Black Carbon emissions. Acidification (from ships); Erosion; Land consumption; Biodiversity loss; Habitat loss/degradation; Waste generation[147]		X	X		X						Ports have a significant impact on habitat loss, both directly through land consumption and the emissions of the port itself, as well as indirectly, through the introduction of shipping with its own impacts (see below) and industrial and recreational terrestrial traffic.[148] Sulphur emissions in harbours are linked to health issues. [149]
		Climate change: increase of extreme weather conditions (including heavier rain and storms) will affect shipping routes; sea level rise will affect ports; greenhouse gas emissions.[150]	X	X	X	X	X						Research on the impact of climate change on shipping showed that major damage and disruption of climate-related hazards has already occurred to ports around the world, and that such impacts are projected to increase over the coming years and decades.[151]

145 https://www.msp-platform.eu/sites/default/files/sector/pdf/mspforbluegrowth_sectorfiche_offshorewind.pdf; https://www.msp-platform.eu/sector-information/offshore-wind-and-fisheries
146 https://www.politico.eu/article/fishermen-offshore-wind-farms-struggle-to-share-sea/
147 https://publications.jrc.ec.europa.eu/repository/bitstream/JRC51275/regno_jrc51275_regno_jrc51275_maritime_external_costs.pdf%5B1%5D.pdf; https://www.transportenvironment.org/what-we-do/shipping-and-environment/shipping-and-climate-change
148 https://publications.jrc.ec.europa.eu/repository/bitstream/JRC51275/regno_jrc51275_regno_jrc51275_maritime_external_costs.pdf%5B1%5D.pdf
149 http://www.emsa.europa.eu/main/air-pollution/sulphur-directive.html
150 https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en; https://www.msp-platform.eu/sector-information/shipping-and-ports; https://link.springer.com/article/10.1057/s41278-018-0114-z
151 https://onlinelibrary.wiley.com/doi/abs/10.1002/wcc.508

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		Interaction with other uses involves conflicts especially with uses requiring fixed installations.[152]	X	X		X	X						Maritime transport and offshore wind will interfere when new offshore wind farms are to be constructed or existing ones expanded – e.g. in areas where maritime traffic is high. Most disputes stem from questions about potential incidents and diversion. [153]
		Knowledge gap in terms of what level of decarbonisation it is possible to achieve using all the available technologies.[154]	X	X		X	X						
	Restructuring of seabed morphology	Physical disturbance to seabed; changes to hydrological conditions; input of contaminants and other substances; pollution; disruption of benthic communities; altering tidal circulation and transport.[155]	X	X			X						As is the case with many activities that involve disturbing the seabed, particularly large-scale dredging projects (such as the planned yet currently defunct Nicaragua Canal) lead to the exposure of long-buried contaminants. These include persistent organic pollutants, which either end up back in the water, or on land with the disposed material.[156]
	Transport/shipping	Air and water pollution (including oil and chemical spills, grey, bilge and black water, ballast water and antifouling paints); Acoustic pollution (including underwater noise); animal collisions; wastewater & solid waste.[157,158,159]	X	X			X						The American lobster, which has been present on board of Europe-bound vessels have eradicated the local lobster in the Oslo fjord.[160]

152 https://www.msp-platform.eu/sector-information/shipping-and-ports
153 https://www.msp-platform.eu/sector-information/transport-and-offshore-wind
154 https://www.mdpi.com/2071-1050/10/7/2243/htm
155 https://moat.cefas.co.uk/uses-of-the-marine-environment/predominant-pressures-exerted-by-human-activities/ ; http://onlinepubs.trb.org/onlinepubs/circulars/ec206.pdf
156https://www.researchgate.net/publication/226993615_Concentration_and_Distribution_of_Persistent_Organic_Pollutants_POPs_in_Harbour_Dredging_Sediments_A_Pilot_Study_Using_a_Statistical_Approach_for_a_More_Efficient_Characterisation_and_Management; http://onlinepubs.trb.org/onlinepubs/circulars/ec206.pdf
157 http://onlinepubs.trb.org/onlinepubs/circulars/ec206.pdf; https://tethys.pnnl.gov/publications/environmental-effects-marine-transportation; https://www.sciencedirect.com/science/article/pii/S0048969719335624
158 https://www.sciencedirect.com/science/article/pii/S0308597X15003590; https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
159 https://publications.jrc.ec.europa.eu/repository/bitstream/JRC51275/regno_jrc51275_regno_jrc51275_maritime_external_costs.pdf%5B1%5D.pdf
160 https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
	Shipbuilding and repair	Metal working operations, surface treatments including paints and coatings, maintenance, repair and noise to the impacts of dismantling and recycling.[161]	X	X	X		X						Many shipbreaking practices are extremely dangerous to both the workers and the environment. Research in Bangladesh and Taiwan has shown proportionally high rates of cancer and resulting mortality due to exposure to these toxins. In India, a study of seawater and sediment samples near a "ship graveyard" of Alang found significantly elevated levels of heavy metals, petroleum hydrocarbons, E. Coli bacteria and E. Faecalis bacteria.[162]
		The shipbuilding industry is also highly energy-intensive, with most of the energy consumption covered by electricity.[163]	X	X			X						
Coastal Tourism and Leisure	**Tourism and leisure infrastructure / Tourism and leisure activities**	Maritime or land-based activities deteriorate the environment, and coastal tourism is highly dependent on the quality of natural ecosystems. [164,165,166,167]	X	X	X	X	X						Various examples of unsustainable tourism eventually damaging the sector itself, include: • Closure of beaches in Haiti and Costa Rica due to raw sewage from hotels • Loss of tourist appeal of Costa del Sol due to overdevelopment
		Climate change related impacts (e.g. erosion, flooding, sea level rise or droughts) will strongly affect coastal areas and significantly affect tourism.[170,171]	X	X	X		X						

161 https://www.oecd.org/sti/ind/46370308.pdf
162 https://ec.europa.eu/environment/integration/research/newsalert/pdf/ship_recycling_reducing_human_and_environmental_impacts_55si_en.pdf
163 https://www.oecd.org/sti/ind/46370308.pdf
164 https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
165 https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
166 https://www.responsibletravel.org/docs/Global_Trends_in_Coastal_Tourism_by_CESD_Jan_08.pdf
167 http://www.ecounion.eu/wp-content/uploads/2019/06/BLUE-TOURISM-STUDY.pdf
170 https://op.europa.eu/en/publication-detail/-/publication/676bbd4a-7dd9-11e9-9f05-01aa75ed71a1/language-en/
171 https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en; http://www.ecounion.eu/wp-content/uploads/2019/06/BLUE-TOURISM-STUDY.pdf

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
		Most economic benefits do not reach the (negatively) affected people or communities. Financial investors are generally outside the region.[172]	X		X		X						• Approximately 90% of coral reefs destroyed in the Caribbean[168] • In the northern Gulf of Mexico, the seasonal hypoxic zone poses serious threats to the gulf's tourism industry, which annually generates 600.000 jobs[169]
		Continued growth will increase the demand for additional infrastructure and services/activities threatening sustainable development.[173]	X	X	X	X	X						
Public Sector	**Military operations**	Implications of climate change, such as challenges to territorial sovereignty, illegal entries or insecurity of spontaneous migrants; insufficient surveillance and safety challenges (e.g. polluting, human trafficking, piracy).[174]	X	X	X	X	X						Forecast impacts of climate change may directly affect Australian Defence's infrastructure and bases. This includes the potential impact of rising sea levels, flooding, storm surges and coastal erosion on bases and training areas. A significant number of key defence facilities are situated at or just above sea level.[175]
		Conflict with other maritime activities as they can obstruct the proper functioning of marine infrastructures considered indispensable for national safety and security (e.g cables, pipelines. Other maritime activities can also be affected by military operations. [176]	X			X	X						For example, there have been concerns that wind turbines could interfere with defence radar or military underwater cables. There may also be negative impacts on optical, radio and

172 https://www.researchgate.net/publication/228464641_Sustainable_Coastal_Tourism_Problems_and_Management_Options
168 https://www.responsibletravel.org/docs/Global_Trends_in_Coastal_Tourism_by_CESD_Jan_08.pdf
169 https://www.oecd-ilibrary.org/economics/the-ocean-economy-in-2030_9789264251724-en
173 https://www.msp-platform.eu/sector-information/coastal-and-maritime-tourism
174 https://www.defense.gouv.fr/content/download/558233/9668077/OBS_Climat%20et%20d%C3%A9fense_201905-RE-Implications%20of%20Climate%20Change%20in%20the%20South%20Pacific%20by%202030%20-%20SPDMM%20Report.pdf
175 https://www.defense.gouv.fr/content/download/558233/9668077/OBS_Climat%20et%20défense_201905-RE-Implications%20of%20Climate%20Change%20in%20the%20South%20Pacific%20by%202030%20-%20SPDMM%20Report.pdf

176 https://www.msp-platform.eu/sector-information/defence-and-other-uses#overlay-context=users/lisa-simone-de-grunt

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
													hydroacoustic observation and the possibilities of veiling.[177]
		Sectoral challenges depend strongly on the region/nation of the marine environment. EU has noted the challenge of planning (due to the secretive nature of defence).[178]	X	X	X	X	X						
	Research, survey and educational activities	Marine sciences are (a) highly cross-sectoral in impact, (b) highly interdisciplinary scientifically, (c) have strong "emerging science" elements, (c) have disproportionately lower visibility in public perception compared to their disproportionately high scientific challenge and societal relevance.[179]	X	X	X		X						
	Land claim	Sea level rise, erosion.[180]	X	X			X						
	Canalisation and other watercourse modifications	Flooding.[181]	X	X	X	X	X						While watercourse modification, including river restoration, are negatively affected by flooding, they also play an important role in flood risk management by supporting the capacity of natural waterways to retain water.[182]
		Larger vessels expected, requiring broadened waterways.[183]	X	X		X	X						

177 https://www.msp-platform.eu/sector-information/defence-and-other-uses#overlay-context=users/lisa-simone-de-grunt
178 https://www.msp-platform.eu/faq/national-defence-and-security
179 http://ec.europa.eu/research/horizon2020/pdf/contributions/post/european_organisations/marine_sciences_-_ifremer_ifm_geomar_noc.pdf
180 https://www.ctc-n.org/sites/www.ctc-n.org/files/resources/land_claim.pdf
181 http://www.ecrr.org/RiverRestoration/Floodriskmanagement/tabid/2615/Default.aspx
182 http://www.ecrr.org/RiverRestoration/Floodriskmanagement/tabid/2615/Default.aspx
183 https://shippingwatch.com/Ports/article10912726.ece

Sectors	Subsectors	Main risks/threats	Affected sustainability dimensions				Geographic dimensions of the risk / threat						Examples of risks
			Economic	Environmental	Social	Governmental	Global	Atlantic Ocean	Pacific Ocean	Arctic Ocean	Southern Ocean	Indian Ocean	
	Coastal defence and flood protection	Rising sea levels affect coastal management, developing frameworks that consider natural systems becomes a challenge.[184]	X	X	X	X	X						Since 2003, € 6 billion were invested in the flood barrier for development in Venice, Italy. Severely delayed, the current estimate is that the barrier will be ready by 2021 or 2022, and will be obsolete in a few decades due to the pace of sea-level rise. Some underwater structures are already suffering from corrosion.[185]
		Overall, challenge of decision-making for long-term, while future is highly uncertain.[186]	X	X	X	X	X						
	Waste removal	Spreading/transmission of pathogens and heavy metals through wastewater.[187]		X			X						See waste sections above.
		Setting up the infrastructure and framework for a circular economy.[188]	X			X	X						
		Absence of infrastructure for appropriate waste management; lack of legislation dealing waste fractions; the absence of any framework for end-of-life product take-back or implementation of extended producer responsibility.[189]	X	X	X	X	X						

Table 4 details the main subsector specific trends in blue economy. The findings in this table have supported the identification of the main global trends, as presented and discussed in Chapter 2.3. While research has been conducted to ensure that the most significant trends are covered, the table should not be considered exhaustive.

Table 4: Main trends of the blue economy sectors

Sectors	Subsectors	Sectoral trends
Non-Living Resources	**Extraction of minerals**	• Building with nature[190] • Deep sea mining[191] • Digital transformation (big data, IoT, robots, drones) -> optimizing machine productivity[192] • Environmental responsibility, reduction of fuel consumption, emissions, corporate social responsibility[193] • Performance based regulation (targets for environmental performance)[194]

	Extraction of salt	• Number of artisanal production areas is declining; loss of landscape quality and cultural heritage[195] • Subsidence and effects on local communities and ecosystems[196]
	Extraction of oil and gas	• O&G companies increasingly diversify their portfolio (e.g. adding renewables)[197] • Digital innovation, big data, including drones, robots, AI[198] • Increased focus on natural gas (LNG) instead of oil[199]
	Offshore structures (incl. submarine cables, pipelines operations and decommissioning of platforms)	• Increasing discussions on whether to remove offshore structures as required by law vs. leaving hard substrate in place to maintain eco-engineered ecosystems. So far, reefs have been created from oil and gas platforms in the United States, Brunei and Malaysia.[200]
	Extraction of water - Desalination	• Minimising discharge, energy costs & environmental impacts[201] • 80 percent reduction in energy use over the past 20 years[202]
	Storage of CO_2 / Carbon sequestration	• Failing of commercially viable CCS[203]
	Marine litter removal	• Cleanup initiatives, e.g. Ocean Cleanup, Australian Marine Debris Initiative, Two Hands Project. [204] • Fishing for litter. [205] • Citizen initiatives, beach cleanup events. [206] • (Remote sensing) monitoring. [207]
	Storage of gas	• Seismic risk analyses. [208]

184 http://ec.europa.eu/environment/life/project/Projects/index.cfm?fuseaction=home.showFile&rep=file&fil=Good_Practice_Guide.pdf
185 https://e360.yale.edu/features/rising-waters-can-a-massive-sea-barrier-save-venice-from-drowning; https://www.washingtonpost.com/world/europe/how-venices-plan-to-protect-itself-from-flooding-became-a-disaster-in-itself/2019/11/19/7e1fe494-09a8-11ea-8054-289aef6e38a3_story.html
186 https://www.subseauk.com/editorialimages/glen%20jewell%20-%20frazer%20nash.pdf
187 https://www.intechopen.com/books/; http://www.fao.org/3/W5367E/w5367e04.htm; https://www.intechopen.com/books/water-quality/impact-of-wastewater-on-surface-water-quality-in-developing-countries-a-case-study-of-south-africa
188 https://www.unido.org/sites/default/files/files/2019-06/UNIDO_Addressing_the_challenge_of_Marine_Plastic_Litter_Using_Circular_Economy_0.pdf
189 https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6466021/

190 https://www.ecoshape.org
191 https://www.nioz.nl/en/valorisation/deep-sea-science-and-technology
192 https://mining.komatsu/blog/details/the-future-of-mining-four-trends-for-tomorrows-success
193 https://mining.komatsu/blog/details/the-future-of-mining-four-trends-for-tomorrows-success
194 https://www.undp.org/content/dam/undp/library/Sustainable%20Development/Extractives/UNDP-MMFSD-LowResolution.pdf
195 https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3225316/
196 https://www.nlog.nl/sites/default/files/update%20subsidence-final%20frisia%20dec03%20openbaar%20v2.pdf
197 https://www.pwc.com/gx/en/ceo-survey/2019/Theme-assets/reports/pwc-2019-ceo-survey-oil-and-gas-report.pdf
198 https://www.pwc.com/gx/en/ceo-survey/2019/Theme-assets/reports/pwc-2019-ceo-survey-oil-and-gas-report.pdf
199 https://oilprice.com/Energy/Energy-General/The-Next-Big-Trend-In-Offshore-Oil-Gas.html
200 https://www.frontiersin.org/articles/10.3389/fmars.2019.00548/full; https://www.sciencedirect.com/science/article/pii/S0964569118304484?via%3Dihub
201 https://www.sciencedirect.com/book/9780128158180/emerging-technologies-for-sustainable-desalination-handbook
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204 https://www.oceanwatch.org.au/latest-news/blog/5-awesome-programs-tackling-ocean-plastics/
205 http://www.fishingforlitter.org.uk/
206 https://www.eea.europa.eu/themes/water/europes-seas-and-coasts/assessments/marine-litterwatch/briefing
207 https://marinedebris.noaa.gov/sites/default/files/Lippiatt%20et%20al%202013.pdf
208 https://www.nam.nl/gas-en-oliewinning/ondergrondse-gasopslag/belang-van-ondergrondse-gasopslag.html

Living Resources	**Fish and shellfish harvesting**	• Marine fisheries in general overexploited. Fish stocks within safe ecological limits is likely to decline. Fishing effort shows increasing trend.[209] • Seafood standards and certification schemes (over 50), focusing mainly on high value species in developed countries such as salmon, cod, tuna, mackerel, shrimp[210] • Discard ban (push towards more selective fishing) • Increased data collection, analysis and visibility tools to assess impacts of climate change. [211] • Gear upgrades for more selective fishing; supply chain innovations (improved logistics). [212] • Global attention to human rights violations. [213]
	Fish and shellfish processing	• New innovative techniques including high pressure processing (HPP), pulsed electric field (PEF) treatment and minimally heat processing combined with microwave (MW) and radio-frequency (RF). Increased focus on food safety, quality, traceability. Innovative packaging solutions (freshness, time-temperature indicators, biodegradable films) [214]
	Marine plant and algae harvesting	• Increasing interest in macroalgae of multinationals such as Unilever, Nestle, Danone and General Mills; certification available since 2017[215] • Research into utilization in livestock feed and potential to reduce CO_2 emissions[216]
	Hunting and collecting for other purposes	• NGOs increasing active with campaigns to stop migratory bird hunting in EU[217] • Recovery of marine mammal populations after hunting bans[218]
	Aquaculture	• More sustainable feed (more sustainable plant-based proteins; less unsustainable animal proteins)[219] • More focus on prevention of diseases; Reduction in the use of antibiotics; focus on sustainability; evolution of aquaculture systems and equipment.[220] • Moving from coastal to offshore (more efficient dilution); Multi-trophic aquaculture (IMTA); integration of renewable energy with aquaculture (multi-use)[221] • Direct buying (consumer-producer); traceability and transparency in the value chain; genetically improved species[222] • Increased use of recirculation systems[223] • Increased use of sensors[224]

[209] https://www.researchgate.net/publication/323418468_Can_we_meet_the_Target_Status_and_future_trends_for_fisheries_sustainability
[210] https://www.unece.org/fileadmin/DAM/trade/agr/meetings/wp.07/2017/8_DEugui_UNCTAD.pdf
[211] https://ec.europa.eu/fisheries/cfp/fishing_rules/discards_en
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[222] https://techcrunch.com/2019/09/05/theres-an-ocean-of-opportunity-for-startups-targeting-the-seafood-industry/
[223] http://www.eurekaselect.com/135856
[224] https://impaqtproject.eu/

	Blue bioeconomy	• Increased use of big data, IoT[225] • Growing demand for algae biomass for a variety of high-value commercial product (e.g. cosmetics, nutraceuticals, pharmaceuticals) and new bio-based applications (biomaterials and energy) [226].
Renewable Energy	**Renewable energy generation**	• Decreasing offshore auction prices; Increasing water depths Increased turbine capacity; Decreasing global level energy costs[227] • Wind and solar dominate renewable growth[228] • Greater durability of turbines[229] • Increased international grid integration Automation, AI weather forecasting, block chain for energy attribute certificates, 3d printing[230] • Increased attention for underwater noise[231]
	Transmission of electricity and communications	
Maritime Transport	**Transport infrastructure**	• Increased labelling and certification: e.g. Environmental Ship Index and EcoPort[232] • Building with Nature [192]
	Restructuring of seabed morphology	• See extraction of minerals
	Transport/shipping	• Automation, IoT, big data[233] • IT applications, increased vessel size, supply chain integration[234]
	Shipbuilding and repair	• 3D design; IoT; virtual reality; robotic welding; automation; sensor communication; data visualization; simulation; cybersecurity[235]
Coastal Tourism and Leisure	**Tourism and leisure infrastructure**	• Increased use of energy, fresh water, and production of waste. Land clearance, habitat degradation, pollution of waterways. Increased demand by North American and European baby boomers; exponential growth of Chinese tourism. Mass tourism and not eco-tourism is prevalent. Some cruise operators are greening their operations.[236]
	Tourism and leisure activities	
Public Sector	**Military operations**	N/A
	esearch, survey and educational activities	Increased monitoring via satellites and remote sensing[237]
	Land claim	• See extraction of minerals
	Canalisation and other watercourse modifications	• See extraction of minerals
	Coastal defence and flood protection	• See extraction of minerals
	Waste removal	• See waste sections above.

[225] http://www.eurekaselect.com/135856
[226] https://op.europa.eu/en/publication-detail/-/publication/79299d10-8a35-11e8-ac6a-01aa75ed71a1
[227] https://www.energy.gov/eere/wind/articles/top-trends-offshore-wind
[228] https://www.renewableenergyworld.com/2019/02/15/6-key-trends-in-sustainable-and-renewable-energy/#gref
[229] https://www.mckinsey.com/business-functions/sustainability/our-insights/winds-of-change-why-offshore-wind-might-be-the-next-big-thing
[230] https://www2.deloitte.com/us/en/insights/industry/power-and-utilities/global-renewable-energy-trends.html
[231] https://www.researchgate.net/publication/242325078_Underwater_noise_emissions_from_offshore_wind_turbines
[232] https://www.ecoports.com/assets/files/common/publications/Annex_1_Good_Practices_Version_2_July_2013.pdf
[233] https://www.greenport.com/news101/Projects-and-Initiatives/smart-and-sustainable-ports
[234] https://www2.deloitte.com/content/dam/Deloitte/cn/Documents/energy-resources/deloitte-cn-er-global-trends-to-2030-en-170104.pdf
[235] https://www.myforesight.my/2018/07/10/technology-trends-in-shipbuilding-ship-repair-towards-industry-4-0/
[236] https://www.responsibletravel.org/docs/Global_Trends_in_Coastal_Tourism_by_CESD_Jan_08.pdf
[237] https://www.sciencedirect.com/topics/earth-and-planetary-sciences/marine-monitoring

Annex 4: Methodology for the critical review of frameworks for sustainability management

The proposed working definition of a blue economy provides a basis for a critical review of existing frameworks and benchmarks. In this section, we provide the methodological steps for the characterisation, prioritisation and critical review of frameworks for sustainability management of a blue economy. From a compilation of 30 frameworks, 15 frameworks were selected for a critical analysis and effectiveness assessment.

Characterisation of frameworks for sustainability management of a blue economy

For the purpose of this study, a sustainability framework is defined as “a supporting structure to measure the status of the sustainability of activities with criteria and/or indicators in order to plan, manage, and evaluate the desired / accepted status of these activities”. This was based on a literature search and online queries for a preliminary selection of relevant sustainability management frameworks, leading to the selection of 30 most relevant frameworks. To identify those frameworks the research team looked at the frameworks commonly accepted in the literature, specifically taking into consideration the inclusion of more than one dimension of sustainability and a balanced representation of the different sectors. The 30 frameworks for sustainability management of a blue economy selected were analysed against specific aspects, based on the methodology explained further below. As a second step, a critical analysis and effectiveness assessment of 15 prioritised frameworks were conducted. Finally, the prioritised frameworks have been submitted to the peer review group for their validation.

Two types of frameworks have been considered: generic frameworks (containing common indicators) on the one hand, and (sub-)sectoral frameworks on the other hand. At least three frameworks have been selected for each sector of the blue economy to ensure that the established as well as emerging subsectors are sufficiently represented. For the “Public sector”, certification schemes or frameworks were not identified but are covered by the generic frameworks.

Methodology to characterise sustainability frameworks

The methodology to characterise the sustainability frameworks consists of applying 12 aspects to review the frameworks, namely, domain, geographic scope, level, user, sector, nature (mandatory or voluntary), dimensions, type of framework (e.g. certification, principles), sectors covered, use of criteria, use of indicators, and data availability (see Table 5: Aspects for reviewing the 30 frameworks).

Table 5: Aspects for reviewing the 30 frameworks

Aspect	Description
Domain	The options for this component are set out by drawing up the following classifications; 'marine', 'coastal' and 'broader than marine and coastal'.
Geographic scope	Indicates the geographical boundaries of any framework. Frameworks can be focused from local to global contexts but can also be centred solely within national borders.
Level	The next facet is 'Level', referring to the level users can apply the framework to, e.g. local, national or a specific sector.
User	Indicates for whom the use of the framework is intended.
Sector	Refers to which overall sector(s) is/are covered in the generic framework or centred on one sector
Voluntary	Indicates if a framework is voluntarily applied or compulsory.
Dimensions	Refers to the consideration of the four sustainability dimensions considered in Chapter 2: environmental, economic, social and governance.
Type	Short description of the type of framework, e.g. certification, principles, etc.
Sector covered	Describes the sectors that are considered in the framework.
Use of criteria	Defines what to measure. In short, the criteria can be considered as an overarching theme within a dimension, e.g. air-pollution.
Indicators	The indicators indicate "how to measure", e.g.SO2 emissions in a geographical area.
Data availability	The indicators rely on data availability and this last characteristic look at the availability of data. This is a general appreciation, Chapter 0 considers it in more details.

A Multi Criteria Analysis (MCA) was conducted to compare different options (in this case frameworks) on the basis of a set of criteria (referred to here as aspects) in a structured way, with the aim to identify and prioritise those frameworks that are most relevant to the overall scope of the study and more specifically to the development of a set of blue

economy sustainability criteria and indicators (Task 4). This decision support method was applied by two groups of experts from the project team, working independently from each other. Similarities were accepted without further discussion; differences were discussed and solved.

Points were attributed to each framework in order to determine which frameworks will be further analysed. To be selected for the effectiveness assessment, the framework must score a minimum of 6 out of 8 points. Points were attributed for the following aspects: domain (marine and coastal environment), sustainability dimensions (environmental, economic, social, governance), use of criteria, use of indicators. These aspects were considered as crucial for the sustainability assessment and specific enough to allow to prioritise the frameworks. Table 6 presents the scoring system which was applied.

Table 6: Aspects for scoring the frameworks

Aspects	Award
Marine	Yes = 1 point
Coastal	Yes = 1 point
Dimensions	1 point per covered dimension
Indicators	Yes = 1 point
Data availability	Yes = 1 point

Outcomes of the characterisation of frameworks

Of the 30 frameworks analysed, 15 scored higher than 6 and were prioritised for the critical analysis of the effectiveness and relevance of frameworks for a blue economy. Exceptionally, frameworks have also been selected (or not) based on further discussion. This is the case for the frameworks "*Sustainability Assessment Framework for Renewable Energy Technology*" and "*Assessment of sustainability indicators for renewable energy technologies*"[238], both scoring five points but selected for either their structure of the framework or the quality of the indicators. In addition, against predefined methodological choices, expert opinion decided not to select the following frameworks that scored more than 6 points: framework of WWF[239], Giurco & Cooper[240] and UNWTO[241]. The latter are either still in development or were deemed to provide insufficient depth.

For the 15 selected frameworks, the following can be observed. As regards to the relevant 'Domain', most of the frameworks have a stronger emphasis on coastal environments (13 references), followed by marine environments (8 references) and broader than marine and coastal (8 references). A total of 4 frameworks[242] consider all three domains. As

[238] Annex 5 – critical review of sustainability frameworks for a blue economy, rows 16 and 17, respectively.
[239] https://d2ouvy59p0dg6k.cloudfront.net/downloads/wwf_marine_briefing_principles_blue_economy.pdf
[240] https://www.sciencedirect.com/science/article/abs/pii/S0892687512000088
[241] UNWTO (2016): Measuring the Sustainability of Tourism (MST)). Retrieved from http://statistics.unwto.org/mst.
[242] Annex 5 – critical review of sustainability frameworks for a blue economy, rows 3, 12, 14 and 15.

for the ‘Geographic scope’, 11 of the 15 frameworks are at global level, while only 4 on regional, 3 on local and 2 within a specific region (EU and Mediterranean). 8 out of 15 frameworks are solely focused on one ‘subsector’.

All 15 frameworks include the ‘Environmental’ and the ‘Social’ dimension of sustainability. 14 frameworks include the ‘Economic/Financial’ dimension of sustainability, while 10 frameworks include the ‘Governance’ dimension. Extending the analysis to the unselected frameworks, this pattern is slightly less pronounced (25 referring to social dimension and 28 to environmental dimension). This is coherent with the outcomes of Chapter 2: in blue economy definitions, the environmental sustainability dimension is the second most prominent after the economic dimension. The focus on social sustainability, however, is more apparent in the frameworks than in the sustainable blue economy definitions. Out of these frameworks, 9 make use of criteria, while 13 use indicators only; 8 of those frameworks make use of both criteria and indicators. The average number of points obtained by the frameworks is 6.5, with only two having the full 8 points, namely “Sustainable Development Goals” and “Indicators for the blue economy”[243].

Critical analysis of the effectiveness and relevance of the frameworks for a blue economy

The 15 selected frameworks were then assessed for their effectiveness and relevance, also by means of a Multi Criteria Analysis. The outcomes of the section above (characterisation of frameworks for sustainability management of a blue economy) served as a starting point. To guide the analysis, the research team has applied another set of predefined aspects. The first category relates to the suitability of the existing methodology in light of the approach proposed above and consists of: (i) Dimensions, (2) Criteria, (3) Indicators, (4) Data. The second category relates to: (5) Reliance on data availability, (6) Transparency, (7) Application in different geographical locations, (8) Ability to identify risks for unsustainability, (9) Potential to address the main threats and trends identified under Task 2.

The criteria for this methodology are elaborated based on the sustainability assessment (SA) methodology from Sala et al. 2015[244], a complex appraisal method, conducted to support decision-making and policy in a broad environmental, economic and social context, which transcends a purely technical/scientific evaluation. It focusses on the systematisation of knowledge on technical/scientific sustainability evaluation, by addressing critical decision-making elements focussed on by domain experts. The criteria have been analysed by two groups of experts from the project team conducted the analysis, working independently from each other. The two groups were provided with clear guidelines and a common questionnaire (Box 1).

[243] Annex 5 – critical review of sustainability frameworks for a blue economy, rows 3 and 7, respectively.
[244] Sala S., Ciuffo B, Nijkamp P. (2015), A systemic framework for sustainability assessment. Ecological Economics, Volume 119, Pages 314-325,

Box 1.
Questions to guide the assessment of the frameworks for sustainable management of a blue economy

- **Question 1: How many of the four dimensions of sustainability are covered by the framework (environmental, economic, social, governance)?**
  = capability of taking into account and balancing the environmental, social, economic and governance dimensions (comprehensiveness).

- **Question 2: How relevant are criteria considered in this framework for task 4 (Develop a set of sustainability criteria for the blue economy)?**
  = suitability of criteria considered. Criteria are defined as "what to measure".

- **Question 3: How relevant are indicators considered in this framework for task 4?**
  = suitability of indicators considered. Indicators are defined as "how to measure."

- **Question 4: What type of data is used in this framework?**
  = The Blue Economy framework will tend to include as much quantitative data as possible.

- **Question 5: How reliant is the framework on data availability?**
  = Application of framework in data-limited cases.

- **Question 6: What is the degree of transparency of the framework?**
  = Is the framework open source: is the coding accessible and is it possible to adapt it (e.g. open source tools/software…)?

- **Question 7: Is the framework applicable in different geographical locations?**
  = Application in different geographical locations.

- **Question 8: What are the types of users?**
  = The framework can be used by different types of users.

- **Question 9: What is the potential of the framework to address main trends/threats/risks?**
  = Possibility of the framework to anticipate current and future developments.

Outcomes for a sustainability management framework for a blue economy

The outcomes of the nine questions (Box 1) are presented below. Note that question 2 (criteria) and question 3 (indicators) are the most relevant questions regarding the elaboration of the blue economy sustainability framework in Task 4. More information on the evaluation of frameworks for a sustainable blue economy can be found in Annex 6: Evaluation of frameworks for a sustainable blue economy. This serves as a reference document for the discussion on the results provided hereafter.

In terms of the sustainability dimensions, the majority (13 out of 15 frameworks) relate to three or more - addressing the environmental, economic and social dimensions. Yet, only three frameworks include the governance dimension, also referred to as the institutional dimension. Governance is a complex and multi-faceted concept which is heterogeneously defined, rendering the identification and use of relevant indicators more difficult. Relevant criteria for the purpose of developing a sustainable blue economy framework was identified in 8 out of the 15, mostly found in sector-specific frameworks. However, 14 out of 15 frameworks were found to contain suitable indicators. These indicators use different types of data, from semi-quantitative to quantitative data. Seven frameworks developed a scoring mechanism to standardise the outcomes, enabling a comparison between different types of indicators. In terms of data availability, much of the data was identified to be retrievable from public sources. Although all frameworks being moderately (6 out of 15) to very (8 out of 15) reliant on data, accessibility is thus not considered to be an issue. Transparency of the methodology was only partial in the majority of frameworks (9 out of 15), meaning that they had transparent methodologies, though their coding mechanism was not open. Some request an independent auditor to collect the data whilst the final assessment is conducted by a certifier (e.g. Marine Stewardship Council certification).

In terms of geographic coverage, most frameworks (13 out of 15) are applicable at several levels. The subsector specific frameworks are often designed to assess the sustainability of a company or subsector anywhere in the world. About half of the frameworks are targeted at governments and policymakers, whilst consumers are not targeted. Only the frameworks from the Marine Stewardship Council (MSC) and of the Aquaculture Stewardship Council (ASC) have a strong link with this stakeholder group. The framework "Observatory of Sustainability of the Algarve Region for Tourism" also considers tourists and local residents as users of the framework.[245] The analysed frameworks vary considerably in their ability to address trends, threats and risks, with only 4 out of 15 that were able to address trends, threats and risks. These include the UN Sustainable Development Goals[246], the Fisheries Performance Indicators[247] and the Sustainability Assessment of Ports[248] and the Observatory of Sustainability of the Algarve Region for Tourism[249]. 4 out of 15 frameworks did not contain indicators to anticipate current and future developments.

[245] Farinha, F., Oliveira, M. J., Silva, E. M., Lança, R., Pinheiro, M. D., & Miguel, C. (2019). Selection process of sustainable indicators for the Algarve region—Observe Project. Sustainability, 11(2), 444.
[246] Sustainable Development Goals (2019). Retrieved from https://www.un.org/sustainabledevelopment/sustainable-development-goals/on 15-11-2019.
[247] Anderson et al. (2015). The Fishery Performance Indicators: A Management Tool for Triple Bottom Line Outcomes, PLoS ONE 10(5): e0122809.
[248] Schipper, C. A., Vreugdenhil, H., & De Jong, M. P. C. (2017). A sustainability assessment of ports and port-city plans: Comparing ambitions with achievements. Transportation Research Part D: Transport and Environment, 57, 84-111.
[249] Farinha, F., Oliveira, M. J., Silva, E. M., Lança, R., Pinheiro, M. D., & Miguel, C. (2019). Selection process of sustainable indicators for the Algarve region—Observe Project. Sustainability, 11(2), 444.

Annex 5: Critical review of sustainability frameworks for a sustainable blue economy.

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Sustainable Development Goals	17 global goals of the UN for the year 2030. It is a blueprint to achieve a better and more sustainable future for all	Predominantly SDG 14 (Life below water). But also SDG 2 (Zero hunger); SDG 8 Decent work and economic growth); SDG 13 (Climate action) etc	Yes	Yes	Yes SDG 14 focused op marine and coastal ecosystems. However, the link with other SDGs is part of the framework.	Global	National	Governments	Cross-sectoral	Yes	Yes	Yes	Yes	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Framework for Processing and Analysing of Maritime Economic Data in Europe	Establishing a stable and reliable system which over the next decade can provide ready-to-use information and figures to monitor the performance of the blue economy	Set of economic indicators describing blue economy sectors in Europe	Yes	Yes	No	EU & Norway	National and sectoral	Governments and knowledge institutes	Cross-sectoral	n.a.	Yes	No	No	No
SUSTAIN	Spreadsheet-based, self-assessment, decision-support tool which gives a numerical value to individual indicators in the sustainabilit	Contains economic, social well-being, environmental quality, governance indicators and a methodology for assessing sustainability based on the	No	Yes	No	EU	National, regional, local	Governments	Cross-sectoral	Yes	Yes	Yes	Yes	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
	y of coastal management	four dimensions												

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
World Bank Sovereign ESG Data Framework 1.0-	The World Bank's Sovereign ESG (Environment, Social and Governance) Data Framework incorporates data relevant to all 17 Sustainable Development Goals, and organizes data into themes the World Bank views as crucial for financial sector representatives to consider when assessing		Yes	No	Yes	Global	National, regional	governments and investors	Cross-sectoral	Yes	Yes, as sub-theme	Yes	Yes	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
	the contribution of investments or policies to sustainable development. The project aims to better align investors' ESG analysis with key sustainable development policy indicators and analysis, increase data transparency, and support private sector investments													

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
	in emerging markets.													

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Indicators for the Blue Economy	Defining a set of key indicators for the Blue Economy in the Mediterranean	Environmental economic indicators for: Energy; Fisheries/Aquaculture; Tourism; Transport; Bioprospecting. Evaluation of challenges in terms of data availability and indicator-based methodology	Yes	Yes	No	Mediterranean	National	Governments, private sector	Cross-sectoral	Yes	Yes	Yes	No	No
Standards and the blue economy	Analysis of the market and performance characteristics of international sustainability standards operating across both the wild catch and	State of Sustainability Initiatives covers a broad range of environmental social and economic assessment criteria for fisheries and aquaculture	Yes	No	No	Global	Sectoral	Governments, private sector and consumers	Cross-sectoral	N.a.	Yes	Yes	Yes	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
	aquaculture sectors													
UN Global Compact	The United Nations Global Compact is a non-binding United Nations pact to encourage businesses worldwide to adopt sustainable and socially responsible policies, and to report on	Predominantly Principles 7 to 9 related to the Environment.	No	No	Yes	Global	Sectoral	Companies	Cross-sectoral	Yes	No	Yes	Yes	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
	their implementation.													
Principles for a Sustainable Blue Economy (WWF)	To fill the gap in shared understanding about what characterizes a sustainable Blue Economy, and to help ensure that the economic development of the ocean contributes to true prosperity,	Universal principles that can be applied to any part of the oceans, seas or coasts, as well as used by any actor involved in the economic development of the sea.	Yes	Yes	No	Global	Global	Public and private sectors	Cross-sectoral	Yes	Yes	Yes	Yes	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
	today and long into the future, WWF has developed a set of "Principles for a Sustainable Blue Economy."													
Sustainability indicators for aquaculture	Indicators of sustainability to assess aquaculture systems	Covers economic, environmental and social criteria and indicators for aquculture	No	Yes	Yes. Also including a case on freshwater aquaculture	local, regional, global	Farm, sectoral	Researchers, farmers, certifying organizations, investors, and policymakers	Living resources	Yes	Yes	Yes	No	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Fishery Performance Indicators	Assess economic, community and ecological sustainability in fisheries	Covers environmental, social and economic indicators and metrics to measure the sustainability of fisheries	Yes	Yes	Yes. Also including a case freshwater fisheries	regional, national	Sectoral	Harvesters, processors, suppliers, retailers, investors, bankers	Living resources	Yes	*Yes*	Yes	No	Yes
Marine Stewardship Council	Independent non-profit organization which sets a standard for sustainable fishing	Contains a set of performance indicators to assess the status of fisheries	Yes	Yes	No	Global	Sectoral	Governments, Fish producers, fish processors, retail and foodservice companies, scientists, conservation groups and	Living resources	Yes	No	Yes	Yes	Yes (child labour)

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
								consumers.						
Aquaculture Stewardship Council	Independent non-profit organisation and labelling organization that establishes protocol on farmed seafood while ensuring sustainable aquaculture	Contains a set of performance indicators to assess the status of aquaculture	Yes	Yes	Yes. Also including freshwater aquaculture	Global	Sectoral	Governments, Aquaculture producers, seafood processors, retail and foodservice companies, scientists, conservation	Living resources	Yes	No	Yes		Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
								groups and consumers						
Renewable energy sustainability	Development of composite index to reveal development stages of renewable energy sustainability	Identification of sustainability factors and corresponding indicators for the four dimensions	Yes	Yes	Yes	Global	National	Governments	Renewable Energy	Yes	*Yes*	*Yes*	*Yes*	*Yes*

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Sustainability Assessment Framework for Renewable Energy Technology	The framework proposes a mechanism to assess the sustainability of renewable energy technologies.	Sectoral assessment for renewable energy	No	No	Yes	Global	Sectoral	building managers and commercial offices	Renewable Energy	Yes	Yes	Yes	No	Yes
Assessment of sustainability indicators for renewable energy technologies	The non-combustion based renewable electricty generation technologies are assessed against a range of sustainability indicators and using data obtained from the litterature.	Indicators are price of generated electricty, greenhouse gas emissions during the full life cycle of the technology, availability of renewable resources, efficiency of energy conversion, land requirements, water consumption	No	No	Yes	Global	Sectoral	Actors in renewable energy sector	Renewable Energy	Yes	Yes	Yes	No	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
		and social impacts.												
Mining and sustainability : asking the right questions	Offers an expanded conceptualisation of minerals sustainability, spanning production, consumption and recycling and connecting social, ecological, technological, economic and governance	Attempts to pursue sustainability in the minerals sector have largely focused on reducing impacts at mining and processing sites. This article suggests to look at sustainability further than those aspects.	Yes	Yes	Yes	Global	Sectoral	Citizens, companies, policy-makers	Non-Living Resources	Yes	Yes	Yes	Yes	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
	domains, across local and global scales.													
An integrated framework for sustainability assessment of seawater desalination	a universal integrated framework to assess the sustainability of desalination technologies was developed.	It can be universally applied to other desalination technologies and/or other countries.	Yes	Yes	No	Global	Sectoral	Desalination companies, governments	Non-Living Resources	Yes	Yes	Yes	No	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Sustainable decommissioning of offshore oil and gas drilling and production facilities	The article provides a SHORT analysis of environmental sustainability on current practices of decommissioning with respect the current legislative and guidelines.	Guidelines are relevant to the extent that current regulations have not yet sufficiently included BASIC environmental dimensions of sustainability. I.e. Guidelines provide areas of consideration in case of decommisioning.	Yes	No	No	Global	Sectoral	Decommission companies Government	Non-Living Resources	N.a.	No (lipservice)	Yes	No	No

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Sustainability assessment of ports	Determine the efficiency and sustainability of each of the case study port plans, relative to other ports.	Evaluation of ports and port-city plans using environmental, social, economic performance indicators	No	Yes	Hinterland	Regional, local	Local	Governments and Port Authority	Maritime Transport	N.a.	Yes	Yes	Yes	Yes
EcoPorts	Environmental initiative of the European port sector	Self Diagnosis Method with a set of indicators to assess the sustainability of ports	No	Yes	No	European	Port sector	Port Authority	Maritime Transport	Yes	No	Yes	Yes	Yes
Environmental Ship Index	Tool for ports in supporting ship-owners to perform better than legally required	Scoring of vessels according to emissions to the air and water	Yes	Yes	Yes	Global	Port sector	Port Authority, shippers and ship owners	Maritime Transport	Yes	No	Yes	No	No

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Clean Shipping Index	Independent and holistic labelling system of vessels' environmental performance	Scoring of vessels according to emissions to the air and water	Yes	Yes	No	Global	Sectoral	Cargo owners, ship owners,	Maritime Transport	Yes	No	Yes	No	No
A concept of a sustainable maritime transportation system	One objective of this document is to identify the various 'imperatives' or goals that must be met to implement an SMTS, and the activities that will need to be undertaken to achieve them	Includes the three dimensions of sustainable development, the economic, environmental and social dimensions, that are seen as equally important	Yes	No	No	Global	Sectoral	Shipping sector broad (ship owners, education centers, port authorities...)	Maritime Transport	Yes	Yes	Yes	Yes	Yes

Sustainable maritime transport: An operational definition	This paper aims to contribute in the discussion for redefining in an operational manner the sustainability concept of the maritime transport sector by following the initial notion of this concept.	The sustainability concept is currently suffering from low credibility as a result of the plethora of definitions and uses which have been launched from a variety of agencies around the globe. The maritime sector is not an exception to this, since there are numerous initiatives within this sector claiming to have a sustainability orientation; however they are diverse in terms of interpretation and implementation of the sustainability principals. Within the maritime	Yes	No	No	Global	Sectoral	Governments and maritime sector regulators	Maritime Transport	N.a.	Yes	Yes	No	Yes

		industry, there is also a trend of narrow defining sustainability, by focusing for example only on energy consumption and air pollution issues. This narrow definition tends to favour the introduction of particular solutions (mainly technological) that may reach one sustainability goal but simultaneously may contradict with another goal.												

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Global Sustainable Tourism Council	Establishes and manages global sustainable standards for tourism industry	Criteria and indicators for tour operators	No	Yes	Yes	Global	National and sectoral	Governments (destination certification) Hotels and tour opperators (Industry criteria)	Coastal Tourism and Leisure	Yes	Yes	Yes	No	Yes
Association of Arctic Expedition Cruise	Managing responsible, environmentally friendly and safe tourism in the Arctic	Series of practical guidelines on clean seas, communities, wildlife etc.	Yes	Yes	Yes. Includes also touristic activities on land	Regional (Arctic)	Sectoral	Tour operators; tourists	Coastal Tourism and Leisure	Yes	No	Yes	Yes	Yes
Sustainable Tourism Indicators	Define a set of indicators for measuring sustainability in tourism.	Covers environmental, social and economic indicators to measure the sustainability of tourism	No	No	Yes, tourism in general	Global	Sectoral (tourism)	Governments and actors of the tourism sector	Coastal Tourism and Leisure	Yes	Yes	Yes	(Indirectly) yes	Yes

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
UNWTO's Measuring Sustainable Tourism (MST) : IS STILL IN PROGRESS, USING AN INITIATIVE.	Measuring Sustainable Tourism (MST) is in progress of becoming a tool to measure sustainability in Tourism. This is done by integrating MST in earlier global Frameworks "Tourism Satellite Accounts (TSA)" and the "System of Environmental-Economic Accounting (SEEA)". MST concerns	The integrated, and statistical nature of the framework will allow for easy comparison, between areas. Covering Social, Environmental, Economic and Cultural Dimensions. However, this framework is work in progress.	Yes	Yes	Yes	Global	Sectoral (tourism)	Stakeholders in the Tourism Industry, Governments	Coastal Tourism and Leisure	Yes	Yes	Yes	Yes	Yes, also Cultural

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
	Economic, Social and Environmental Dimensions.													
The European Tourism Indicator System ETIS toolkit for sustainable destination management (2016)	The specific objective of the ETIS is to contribute to improving the sustainable management of destinations. It aims at helping	Covers environmental, social (and cultural) and economic indicators to measure the sustainability of tourism.	No	Yes	Yes	Global	Local and sectoral	All actors in the Tourism industry	Coastal Tourism and Leisure	Yes	Yes	Yes	No	Yes, also Cultural

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
	destinations and the stakeholders within to measure their sustainability management processes, enabling them to monitor their performance and progress over time.													

Framework	Short description	Relevance	DOMAIN			Geographic scope	Level	User	Sector	Voluntary	Dimensions			
			Marine	Coastal	Broader than marine and coastal						Economic/Financial	Environmental	Governance	Social
Observatory of Sustainability of the Algarve Region for Tourism(OBSERVE) (2019)	The specific objective of the OBSERVE (2019) project has been the development of indicators for SD in Tourism in the Algarve Region, which are relevant for stakeholders (and comply with criteria, e.g. data availability)	Framework covers environmental, Sociocultural, Economic and Institutional indicators to measure the sustainability of tourism in the region.	No	Yes	No	Local/ Regional	Regional (local) and sectoral	All stakeholders in the region (both public, private and consumers - touristis and local residents)	Coastal tourism and Leisure	Yes	Yes	Yes	Yes	Yes

Framework	Type	Sectors covered	Use of criteria	Use of indicators	Data availability	Suggestions for BEDF (task 4)	Selected for	Argument	Points (marine (yes=1 point), coastal (yes=1 point), 4 dimensions (1 dimension = 1 point), indicators (yes = 1 point), data availability (yes = 1 point). Minimum for selection is 6. 5 may be included, based on arguments of the team for the selection.	Source
Sustainable Development Goals	Guidelines	marine pollution ocean acidification Fishing / aquaculture tourism scientific knowledge, research capacity and transfer marine technology conservation	Yes, 10 targets	Yes	Yes except for some indicators that can not be measured	All targets of SDG14 will be considered for the BEDF.	Yes	Thethe data under SDGs will be mainly at national level but SDGs are a internationally accepted guideline which should be refle	8	UN (2019). Sustainable Development Goals. Retrieved from https://www.un.org/sustainabledevelopment/sustainable-development-goals/on 15-11-2019.

								cted in our BED F.		
Framework for Processing and Analysing of Maritime Economic Data in Europe	Framework	EC sectors	No	Yes	Yes NACE, Eurostat, DCF/JRC, COFOG	Use method for GVA and employment The framework is useful for the indicator information and it constitutes European Blue Economy database. It is however not the type of framework we develop in Task 4 and it therefore does not qualify for task 3.2. but we will keep an eye on the useful approach for indicator information.	No	Study introduced method to measure the performance of the blue economy. Not to assess the	5	COGEA (2017). Study on the Establishment of a Framework for Processing and Analysing of Maritime Economic Data in Europe. Publications Office of the European Union. Luxembourg. Retrieved from https://op.europa.eu/en/publication-detail/-/publication/9c132514-982d-11e7-b92d-01aa75ed71a1 on 22-11-2019.

								sustainability.		
SUSTAIN	Ddecision-support tool	Not applicable to specific sectors	No	Yes	Yes. Framework built on commonly used data and data from official monitoring in the EU	Use framework for task 4. The structure and methodology are very relevant but will have to be tailored to BE.	Yes	Although man indicators may not be relevant, the framework is.	7	Sustain (2019). Measuring Coastal Sustainability: The sustain policy tool. Retrieved fromhttp://www.sustain-eu.net/ on 15-11-2019.

World Bank Sovereign ESG Data Framework 1.0-	Platform for data	Emissions, natural capital for freshwater, marine protected areas, renewable energy	Yes	No	Yes	Similar rationale than for Task 4. Use it also for the data base	No	Not specific to Blue Economy	5	Kurtz, T. (2019). World Bank Blogs: New resources for sovereign ESG data and investors. Retrieved from https://blogs.worldbank.org/opendata/new-resources-sovereign-esg-data-and-investors?cid=ECR_LI_WorldBank_EN_EXT#3 on 22-11-2019.

Indicators for the Blue Economy	Structured indicators	Energy, Fisheries, Tourism, Maritime transport, Biopros pection	No	Yes	Yes	Recommendation: national data need to be provided according to common standards. The indicators considered per sector can also be used in Task 4. Their selection follows RACER. This study focusses on Med Sea, including a number of indicators (economic & env). It is mostly relevant if we need Med Sea specific indicators in Task 5 or 6.	Yes	Clear structure of limited indicators and multisectoral	8	Giraud, J., Lafitte, A. & Fosse, J. (2017). Blue economy; economic activities and sustainable development. Plan Blue. Notes 34. Valbonne, France. Retrieved from https://planbleu.org/sites/default/files/publications/note_34_en_web.pdf on 22-11-2029.
Standards and the blue economy	Review (9 certifications)	Seafood industry	No	No	No		No	It is a review and not a framework.	5	Potts, J., Wilkings, A., Lynch, M., & MacFatridge, S. (2016). State of Sustainability Initiatives Review: Standards and the blue economy. International Institute for Sustainable Development. Retrieved from https://unctad.org/meetings/en/Contribution/ditc-ted-oceans-ssi-blue-economy-

										2016.pdf on 22-11-2019.
UN Global Compact	Principles	All sector	Yes, guidelines	No	n.a.	Consider principle 7 to 9 for industries. UN Global Compact is an internationally accepted guideline which should be reflected in our BEDF.	No	The framework provides guidelines but not really keys for an assessment of sustainability	3	UN Global Compact (2019). The power of principles. Retrieved from https://www.unglobalcompact.org/what-is-gc/mission/principles on 15-11-2019.

Principles for a Sustainable Blue Economy (WWF)	Principles	All actors of the Blue Economy, including governments, private and financial sector actors, international agencies, and civil society groups.	No	No	No	Make sure our indicators cover those principles. These principles are more relvant to the definition - Task 2 - than to the framework development, to make sure that the proposed defitions is in line with the universal principles for SUSTAINABLE BLUE ECONOMY	No	The framework provides guidelines but not really keys for an assesment of sustainability	6	WWF (2018). Principles for a Sustainable BLUE ECONOMY.Retrieved from https://d2ouvy59p0dg6k.cloudfront.net/downloads/wwf_marine_briefing_principles_blue_economy.pdf on 15-11-2019.
Sustainability indicators for aquaculture	Structured indicators	Aquaculture	Yes	Yes	Data comes from semi-structured interviews, observations, websites	Several indicators maybe relevant for task 4	Yes	Article contains potentially relevant indicators for assessment of aqu	6	Valenti, W. C., Kimpara, J. M., Preto, B. D. L., & Moraes-Valenti, P. (2018). Indicators of sustainability to assess aquaculture systems. Ecological indicators, 88, 402-413.

								acult ure		
Fisher y Perfor mance Indica tors	Structur ed indicator s	Fisherie s	No	Yes	Focus on data poor cases	Several indicators maybe relevant for task 4	Y e s	Artic le cont ains pote ntiall y relev ant indic ator s for asse ssm ent of fishe ries	6	Anderson J.L., Anderson C.M., Chu J., Meredith J., Asche F., Sylvia G., Smith M. D., Anggraeni, D., Arthur, B., Guttormsen, A., McCluney, J. K., Ward, T., Akpalu, W., Eggert, H., Flores, J., Freeman, M. A., Holland, D. S., Knapp, G., Kobayashi, M., Larkin, S., MacLauchlin, K., Schnier, K., Soboil, M., Tveteras, S., Uchida, H. & Valderrama, D. (2015). The Fishery Performance Indicators: A Management Tool for Triple Bottom Line Outcomes. PLoS ONE 10(5): e0122809.

Marine Stewardship Council	Scheme/label/certification	Fisheries	No	Yes (28 performance indicators)	Yes, data on stock assessments, many other types of data	Screen for useful indicators for fisheries	Yes	Detailed indicators on fisheries. The framework however focuses on single sector; coverage of multi sectors preferred.	6	MSC (2019). Get Certified Your guide to the MSC fishery assessment process. Retrieved from https://www.msc.org/docs/default-source/default-document-library/for-business/for-fishery-clients/fisheries-get-certified-2019.pdf on 18-11-2019.

Aquaculture Stewardship Council	Scheme/label/certification	Aquaculture	Yes	Yes	Farm specific data; farm traceability data	Screen for useful indicators for fisheries	Yes	Detailed indicators on aquaculture. The framework however focuses on single sector; coverage of multi sectors preferred.	6	ASC (2019). Certifying environmentally and socially responsible seafood. Retrieved from https://www.asc-aqua.org/ on 15-11-2019.
Renewable energy sustainability	Index calculating a z-score	Energy sector	Yes	No	Yes, leading to a score	The way factors are structured may be relevant for the EBDF. It's not a framework in the sense of Task 4 but offers some goals that IMO is working towards. The score allows for a scale in sustainability	Yes	The index calculates a score whic	7	Cîrstea, S., Moldovan-Teselios, C., Cîrstea, A., Turcu, A., & Darab, C. (2018). Evaluating renewable energy sustainability by composite index.

						(more or less succesul sustainable cases)		h allo ws nua nces in the sust aina bility scal e. How ever , the inde x com pare s coun tries, not sect ors.		Sustainability, 10(3), 811.
Sustai nabilit y Asses sment Frame work for Rene wable Energ y Techn ology	Sustain ability assess ment	renewa ble energy sector	Ye s, call ed fac tor s	Yes	Yes, data provid ed by the user	None	Y e s	Stru ctur e of the fram ewor k	5	Luong, S., Liu, K., & Robey, J. (2012). Sustainability assessment framework for renewable energy technology. Technol. Sustain. Built Environ. Cent, 1-8. Retrieved from https://www.readin g.ac.uk/web/files/ts

										be/Luong_TSBE_Conference_Paper_2012.pdf on 22-11-2019.
Assessment of sustainability indicators for renewable energy technologies	Sustainability assessment	renewable energy sector	No	Yes	From different sources per indicator	Consider the relevance of some indicators for the framework	Yes	Clear measurable indicators - not the structure of the framework	5	Evans, A., Strezov, V., & Evans, T. J. (2009). Assessment of sustainability indicators for renewable energy technologies. Renewable and sustainable energy reviews, 13(5), 1082-1088. Retrieved from https://relooney.com/NS4053-Energy/0-Harold_18.pdf on 22-11-2019.

Mining and sustainability: asking the right questions	Sustainability assessment	Mining sectors	No	Yes	No	Add a 5th dimension: technological dimension?	No		7	Giurco, D., & Cooper, C. (2012). Mining and sustainability: asking the right questions. Minerals Engineering, 29, 3-12. Retrieved from https://www.sciencedirect.com/science/article/abs/pii/S0892687512000088 on 22-11-2019.
An integrated framework for sustainability assessment of seawater desalination	Sustainability assessment	Water extraction/desalinisation sector	Yes	Yes	Yes	See for desalinisation in Kuwait: https://www.sciencedirect.com/science/article/abs/pii/S0011916416318690	Yes		7	Ibrahim, Y., Arafat, H. A., Mezher, T., & AlMarzooqi, F. (2018). An integrated framework for sustainability assessment of seawater desalination. Desalination, 447, 1-17. Retrieved from https://www.sciencedirect.com/science/article/abs/pii/S0011916418310841 on 22-11-2019.

Sustainable decommissioning of offshore oil and gas drilling and production facilities	Case-studies leading to recommendations	Oil and Gas extraction (Decomisioning of offshore facilities), Fisheries (partially).	Yes	No	No	This article provides some criteria to consider (relevant), but no indicators	No	The article provides a critical view on current practices and case studies + suggestions for sustainability. However there is no real indication on keys for an asse	3	Khan, M. I., Smit, E., Lakhal, S. Y., & Islam, M. R. (2006). Sustainable decommissioning of offshore oil and gas drilling and production facilities. In The 36th CIE Conference on Computers & Industrial Engineering (pp. 4357-4373). Retrieved from https://www.researchgate.net/publication/283251718_Sustainable_decommissioning_of_offshore_oil_and_gas_drilling_and_production_facilities on 22-11-2019.

								sment of sustainability		
Sustainability assessment of ports	Sustainability assessment	Ports, shipping, fisheries, tourism	No	Yes, see dimensions	Location-specific information accessible from global databases, generic standardized data, and compliance of regula	Some indicators for ports maybe relevant. SICI could also be relevant.	Yes	Article contains potentially relevant indicators for assessment scale of ports	7	Schipper, C. A., Vreugdenhil, H., & De Jong, M. P. C. (2017). A sustainability assessment of ports and port-city plans: Comparing ambitions with achievements. Transportation Research Part D: Transport and Environment, 57, 84-111.

					tions and guidelines.					
EcoPorts	Certification	Ports, shipping	Yes	Yes	No		No		5	Ecoports (2019). Retrieved from https://www.ecoports.com on 22-11-2019.
Environmental Ship Index	Index calculating a score	Shipping	No	Yes	Yes	Giving a score is an interesting way to scale sustainability	No	This framework does not take enough aspects into account	5	WPSP (2019). Environmental Ship Index. Retrieved from https://www.environmentalshipindex.org/Public/Home on 22-11-2019.
Clean Shipping Index	Scheme/label/certification	Shipping	No	No. Instead use of par	Data required on SOx, NOx, PM,	Scoring mechanism could be useful.	No	Suggest to use more	3	Clean Shipping Index (2019). Retrieved from https://www.cleans

				ameters.	CO2, chemicals, waste water			generic indicators applicable to all sectors		hippingindex.com/ on 22-11-2019.
A concept of a sustainable maritime transportation system	Goals	Shipping sectors broad	Yes	Yes	Yes but not all activities/indicators are measurable	Make sure our indicators cover those goals	Yes	Offers a broad perspective on sustainability. It may be make the scope too large but some activities can	7	IMO (2019). World Maritime Day: a concept of sustainable maritime transportation system. London, United Kingdom.Retrieved from http://www.imo.org/en/About/Events/WorldMaritimeDay/WMD2013/Documents/CONCEPT%20OF%20%20SUSTAINABLE%20MARITIME%20TRANSPORT%20SYSTEM.pdf 22-11-2019.

								be relevant for the final framework		

Sustainable maritime transport: An operational definition	Analysis of definition and refelction on available tools	Maritime transport sector	No	No	No	This overview paper provides a lot of useful information on different sustainability frameworks. Not for Task 3.2 but keep on the radar. Definitions of these two terms are provided below. The SS regards natural capital as providing some functions that are not substitutable by man-made (produced) capital and should be maintained. The WS regards that man-made (produced) capital of equal value can take the place of natural capital. A general delineation between the SS and the WS as has been adopted from is presented in the following Table 1. When selecting indicators for measuring the performance of any system it is important to avoid confusing goals and objectives. Goals are what society ultimately wants. Objectives are things	No	The article is mainly a reflection on definitions and tools but does not provide one framework as such. However, the article is useful to reflect on important aspects for the	4	Chatzinikolaou, S. & Ventikos, N. (2011). Sustainable maritime transport: An operational definition. In Sustainable Maritime Transportation and Exploitation of Sea Resources (Vol. 931, No. 939, pp. 931-939). Routledge in association with GSE Research. Retrieved from https://www.researchgate.net/publication/280313304_Sustainable_maritime_transport_An_operational_definition on 22-11-2019.

						that help achieve goals, but are not ends in themselves and indicators are variables selected and defined to measure progress toward an objective		fram ewor k.		

Global Sustainable Tourism Council	Certification	Tourism industry / private sector travel industry, focusing currently on Hotels and Tour Operator	Yes	Yes	No	This framework can be relevant if indicators are developped per sector - for coastal tourism	No		5	GSTC (2019). GSTC 2019 Global Conference. Retrieved from https://www.gstcouncil.org on 15-11-2019.
Association of Arctic Expedition Cruise	Scheme /label/certification	Tourism industry / private sector travel industry	No	No	n.a.	Framework uses guidelines. Therefore not useful for task 4	No	Framework uses guidelines. Therefore not useful for task 4	5	AECO (2019). Guidelines. Retrieved from https://www.aeco.no/guidelines/on 15-11-2019.

Sustainable Tourism Indicators	System of sustainability Indicators	Tourism industry / private sector travel industry	Yes	Yes	No	None	No	The framework is not selected but some concrete indicators for the sector may still be relevant to use in the framework for the Blue Economy.	5	Guerreiro, S. & Seguro, P. (2018) Sustainable Tourism Indicators. Retrieved from https://bit.ly/32Jmuty on 15-11-2019.

UNWTO's Measuring Sustainable Tourism (MST): IS STILL IN PROGRESS, USING AN INITIATIVE.	Indicator System for management tool	Tourism industry / private sector	Yes	Yes	Yes	The Economic, Environmental and Social dimensions are very emphasized and clearly developed. It is build on other frameworks and it is measuring in flows (indicators for all activities)	No	This framework is work in progress and can therefore not qualify for task 3.2.	8	UNWTO (2016). Measuring the Sustainability of Tourism (MST). Retrieved from http://statistics.unwto.org/mst on 15-11-2019.
The European Tourism Indicator System ETIS toolkit for sustainable destination management (2016)	Indicator System for management tool	Tourism industry / private sector / Energy / Waste Maneg ment	Yes	Yes	Yes, should be availlable from "each SWG member's sector, employment role and interest or be acces	Criteria and Indicators , p.21 onwards	Yes	The Economic, Environmental and Social dimensions are very emphasized and	6	European Commision (2016). The European Tourism Indicator System: ETIS toolkit for sustainable destination management. Publications Office of the European Union. Luxembourg. Retrieved from https://ec.europa.eu/docsroom/documents/21749 on 22-11-2019.

					sible through other authorities, e.g. at a national level".			clearly developed. Very specific, multiple dimensions.		
Observatory of Sustainability of the Algarve Region for Tourism(OBSERVE) (2019)	Indicator System for management tool	Tourism industry / Private sector / Energy / Waste Maneg ment	Yes	Yes	Yes, but in a lot of cases from local/regional databases.	Some of the criteria and indicators could be usefull (p. 11).	Yes	The framework covers the four sustainability dimensions, provides an interesting	7	Farinha, F., Oliveira, M. J., Silva, E. M., Lança, R., Pinheiro, M. D., & Miguel, C. (2019). Selection process of sustainable indicators for the Algarve region—Observe Project. *Sustainability*, *11*(2), 444.

								structure and relevant criteria and indicators for the tourism sector in a coastal region. Furthermore the relevance of indicators and criteria have been teste		

								d on stak ehol ders .		

Annex 6: Evaluation of frameworks for a sustainable blue economy

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code [250]	Comment	Code [251]		Code [252]	Comment	Code [253]	Comment	Code [254]	Comment	Code [255]	Comment
Sustainable Development Goals	D	Dimensions not explicit. Links to environment, governan	A	No Criteria, only Targets	C	Sustainable fisheries and marine litter may include	C	% and descriptive	C	Most targets require specific data that needs to be	D	Open framework

[250] Q1: A/ None; B/ 1; C/ 2; D/ 3 or 4
[251] Q2: A/ Criteria do not exist, B/ Criteria are not suitable for BE framework, C/ Some criteria are suitable, D/ Most criteria are suitable
[252] Q3: A/ Indicators do not exist; B/ Indicators are not suitable for BE framework; C/ Some indicators are suitable; D/ Most indicators are suitable
[253] Q4: A/ No data; B/ Qualitative / relative data; C/ Semi-quantitative; D/ Quantitative
[254] Q5: A/ Not specified; B/ Very reliant on data availability; C/ Moderately reliant on data availability; D/ Not reliant on data availability
[255] A/ Not specified; B/ Closed framework; C/ Partially open framework; D/ Open framework / transparent values

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
		ce, economic, social.				relevant indicators				collected via national statics agencies.		
SUSTAIN	D	Environmental, economical, social, governance	A	No Criteria, only issues.	D	Multiple indicators may be suitable.	C	Mostly semi-quantitative for environmental, social and economic dimensions. Quantitative: %, number of days, tons per capita…	C	3 dimensions are dependent on data availability. Only governance can be done with expert opinion.	C	The framework is publicly open to be used but not to be modified or adapted. This last action requires a password.

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
Indicators for the Blue Economy	C	Economic and environmental	A	N.a.	D	Limited number of indicators that relate to the Blue Economy and cover different sectors	D	%, added value(€) and numerical values	B	All indicators rely on data availability	C	Unclear how dashboard is produced.

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
Sustainability indicators for aquaculture	D	Social, economic, environmental	C	Most criteria are relevant for task 4	C	Interesting indicators may be: Pollution Resilience Capacity for expansion Use of water and energy Risk to biodiversity Risk to employees Inclusion	D	Mixture of data is required, ranging from percentages to numerical values.	B	The indicators require very specific mostly quantitative data	D	The approach of the framework is fully open and transparent

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
Fishery Performance Indicators	D	Social, economic, environmental	C	Dimensions are seen as criteria and some are relevant	C	In this framework metrics can be interpreted as indicators	D	Mixture of data is required, but mostly quantitative and sometimes relative.	C	The FPIs are able to assess data-poor fisheries—and sectors—with sufficient confidence and consistency to capture meaningful variation in performa	D	Open framework that allows fisheries to make use of. Open sourced.

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
										nce among fisheries.		

Marine Stewardship Council	D	Environmental and governance (child labour covers the social aspect)	D	Relevant criteria can be: Stock evaluation Impact on endangered, threatened or protected (ETP) species Impact on habitats Impact on the ecosystem Governance and policy Fishery specific management system	D	Some indicators related to the criteria mentioned in question 2 can be considered for the BE framework. The bullet points under criteria are seen as indicators.	C	A fishery assessment is based on expert analysis of information (yes/no answers).	B	The client must provide enough information and data. Fisheries without available data can use MSC's Risk-Based Framework.	C	All the data is reported by a fishing company to the MSC. The MSC determines if it is correct and if a company is eligible for a certification. Access to the indicators but not to the coding.

Aquaculture Stewardship Council	C	Environmental, social	C	Possibly interesting criteria: Criterion 1.1 Compliance with all applicable local, national and international legal and regulatory requirements Criterion 2.1 Benthic biodiversity and benthic effects Criterion 2.2. Water quality in and near the site of operations	C	Some indicators related to the criteria mentioned in question 2 can be considered for the BE framework	B	Mostly qualitative data combined with semi-quantitative data	C	The client must provide enough information and data, completed by expert judgement	C	All the data is reported by a fishing company to the ASC. The ASC determines if it is correct and if a company is eligible for a certification. Access to the indicators but not to the coding.

				Criterion 3.2. Introduction of transgenic species Criterion 5.3 Environmental welfare Criterion 6.1. Child labour and young workers Criterion 6.4 Work environment health and safety Criterion 7.1. Community engagement and effective conflict resolution								

				Criterion 7.2. Respect for local cultures and traditional territories Criterion 7.3. Access to resources								

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
Renewable energy sustainability	D	Environmental, social, economic. Governance is called 'Institutional'.	C	Criteria are called Factors. Some are relevant: Environmental sustainability Innovation competitiveness	C	Some indicators related to the criteria mentioned in question 2 can be considered for the BE framework.	C	Ranking, %, Score 1-7.	B	The framework is reliant on data but location for data is provided and can be obtained by anyone or any organization.	C	Data can be obtained from a designated institute. The formula on establishing the end score is already determined and cannot be altered.

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
Sustainability Assessment Framework for Renewable Energy Technology	D	Economic, social, environment and technical	C	Criteria are called Factors. Some are relevant.	C	Indicators are very specific to the sector. Limited applicability. Potentially relevant indicators: aesthetic impact, GHGs, costs, taxes, subsidies.	D	Predominantly quantitative. In some cases qualitative/semi-quantitative	B	Specifically reliant on economic data	C	Every indicator and aspect is open and can be assessed for every organization themself. The framework is open but methodology for calculating indicators is not clear in all cases

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
Assessment of sustainability indicators for renewable energy technologies	D	Social, environmental, economic, technical	A	n.a.	C	The indicators are very generic. There are 7 indicators: Price ,CO2 Emissions, Availability and limitations, Efficiency ,Land use, Water consumption and	C	A ranking from 1 to 4. Some indicators are quantitative.	C	Data is necessary but easily accessible .	C	Used methodology is not clear. Values/thresholds are not provided. Also the kind of indicators is sometimes unclear.

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
						Social impact.						

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
An integrated framework for sustainability assessment of seawater desalination	D	Environmental, techno-economic, social	A	n.a.	C	Most of the environmental indicators could be suitable, e.g. GHGs. For social indicators, employment and equity maybe used.	D	Quantitative data is used combined with expert opinion. Values are normalised/standardised in order to compare different units of measurement.	C	Some indicators can be assessed using expert opinion. Most other indicators need data.	C	The decision-making framework provides transparent values and can be altered by its user. The weighing system is rather complex.

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
Sustainability assessment of ports	D	Social, environmental, economic	A	n.a.	C	Social: Well-being, human rights, climate robustness, urbanization, safety against floods. Environment: erosion, sedimentation, maintenance	C	Semi-quantitative and quantitative data is used, combined with expert opinion. Values are normalised/standardised in order to compare different units of measurement.	B	Reliant on publicly available data from comprehensive studies to compare with qualitative description of sustainability plan	D	Assessment is fully transparent and replicable by others

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
						dredging, sediment quality, ballast water treatment. Social: hinterland connections.						

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
A concept of a sustainable maritime transportation system	D	Economic, social and environmental	A	Only overall goals are mentioned	A	Only activities that contribute to the goals	A	It makes statements about promotion, continuation and implementation of programs, policies etc.	A	Information can be obtained from literature review. Specific data is not required.	C	The activities and the goals are transparent. IMO functions as the single coordinator. The activities and the goals are transparent. IMO functions as the single coordinator.

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
Observatory of Sustainability of the Algarve Region for Tourism (OBSERVE)(2019)	D	Environmental, sociocultural, economic and institutional.	D	Referred to as Sub-Domains; the article suggests many criteria that they have tested on stakeholders. Relevant criteria can be: Climate and Climate Change,	C	Energy usage, water usage, tourism intensity, Nº beaches and marinas with blue flag , Nº embarked and disembarked passengers of cruise ships	D	%, total amount, etc.	B	Quantitative data required, but most data free available in national data base.	D	Transparent and no advanced methodology behind framework

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
				environmental management, energy management, water cycle management, materials and waste management, natural capital management, territory		(Portimão Port), Electricity consumption per inhabitant, Tourists who repeat their visit to Portugal,						

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
				management, pressure, demography, social cohesion								
The European Tourism Indicator System ETIS toolkit for sustainable destination	D	Destination management, economic value, social and cultural impact,	C	Social: Protecting and enhancing cultural heritage, local identity and assets.	C	Economic: specific for the tourism sector. Social: contains a lot of additional	C	Numerical values and percentages. %, km, tonnage	B	Mostly moderately dependent on data availability. Mostly quantitative data is	D	Can be filled in on a voluntary basis. Factsheet and toolkits freely available online.

Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
management (2016)		environmental impact		Environmental: Reducing transport impact; Climate Change; Solid waste management; Sewage. treatment ;Landscape and biodiversity protection.		indicators for (already) established indicators in LOT2. Environmental: sector specific indicators.				required. Expert opinion can also be sufficient in some cases.		Verified by a third party.

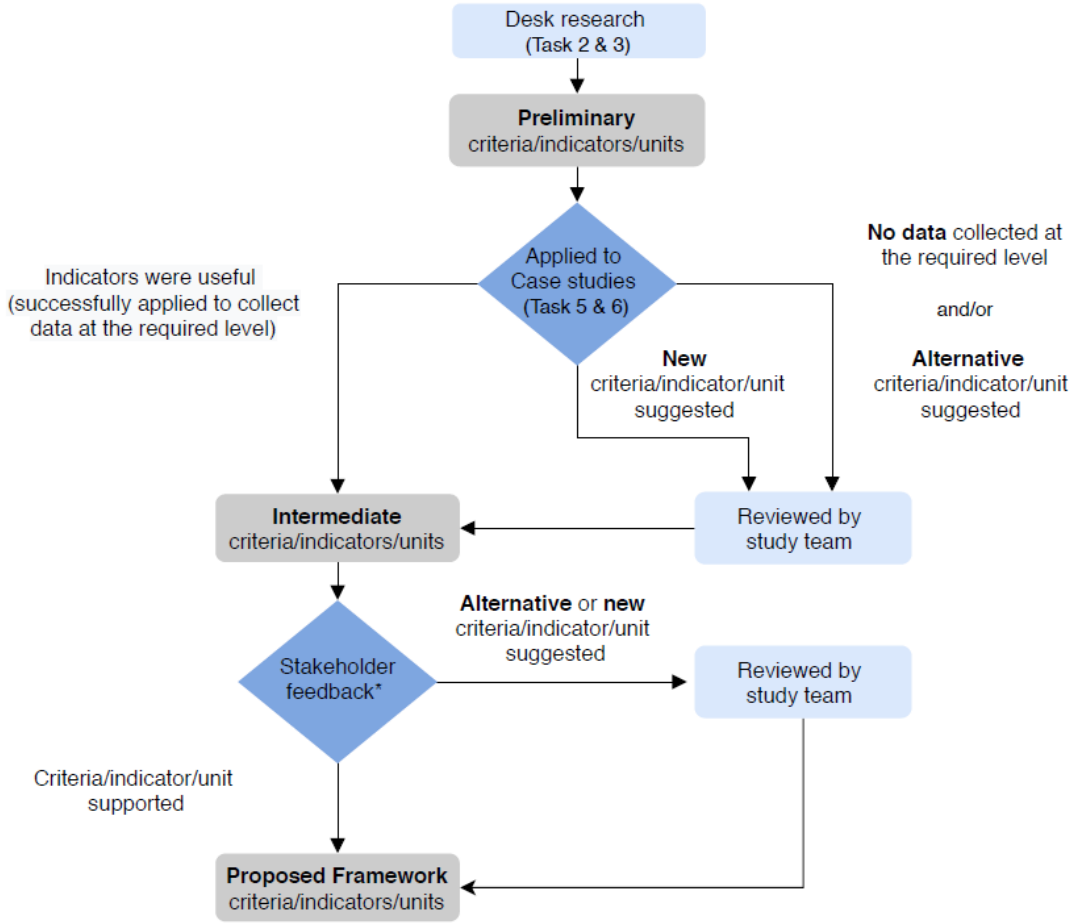
Dimensions / Framework	1. How many of the four dimensions are covered by the framework (environmental, economical, social, governance) ?		2. How relevant are criteria considered in this framework for task 4 ?		3. How relevant are indicators considered in this framework for task 4 ?		4. What type of data is used in this framework ?		5. How reliant is the framework on data availability ?		6. What is the degree of transparency of the framework?	
	Code 250	Comment	Code 251		Code 252	Comment	Code 253	Comment	Code 254	Comment	Code 255	Comment
				Economic: Tourism supply chain.								

Annex 7: Methodology for the development of a set of sustainability criteria and indicators

This annex provides the methodological steps undertaken to develop and validate a set of sustainability criteria and indicators for a blue economy, building on the established and emerging sectors identified of Task 2 (see Annex 1, 2 and 3), as well as trends shaping and affecting the blue economy. The proposed working definition (see main report) provides a clear, high-level statement for the vision of a sustainable blue economy.

A similar development process has been followed for both the common (generic), subsector-specific and operational criteria and indicators. The set of indicators developed for the Blue Economy Sustainability Framework underwent several processes for review and fine-tuning. First and foremost, the indicator set has been tested on various case studies (see Chapter 5 and 6 of the report). It also underwent an expert review. Figure 2 describes the full process of refining the indicators through the case studies and expert consultation.

Figure 2: Criteria and indicator development flowchart



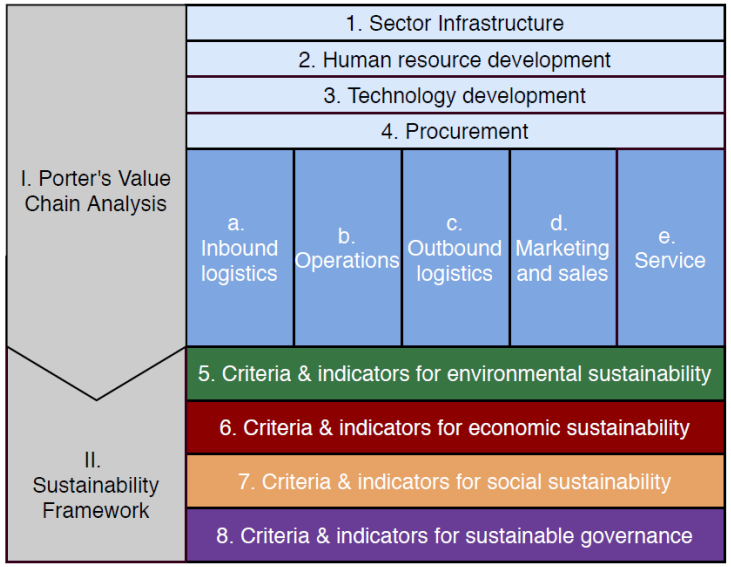
**Sectoral experts, European Commission and other relevant stakeholders*

Initial desk research: Porter's Value Chain Analysis (VCA)

The research team applied an analytical framework to develop a set of sustainability criteria and indicators for the Blue Economy, firstly based on the Porters' value chain approach (VCA). The VCA provides a horizontal qualitative narrative of the sustainability aspects in all the segments of the value chain of a blue economy sector. Then a sustainability framework applies a set of indicators to each of the four dimensions for a vertical, in-depth and quantitative approach, focussing on the operational part of the value chain.

The proposed sustainable blue economy framework uses the structure and approach of the SUSTAIN framework[256] and integrates relevant criteria and indicators derived from the outcomes of Task 3. Through the proposed methodology, generic sustainability issues of a sector can be traced to specific segments of the value chain.

Figure 3: the blue economy sustainability framework approach



Source: own elaboration

The objective of the Value Chain Analysis approach (VCA) is to introduce and characterise a blue economy sector. The theoretical background originates from Porter (1985): a value chain is defined as "a set of activities that a business carries out to create value for its customers". It is useful to examine the business activities; see how they are interconnected and understand which of them add value (and so identify sources of competitive advantage)[257]. Porter's VCA enables the analysis of sectors distinguishing the

[256] SUSTAIN was one of the frameworks assessment: an indicator-based methodology and scoring system which enables a self- assessment approach for local and regional authorities, to evaluate their sustainability performance for the purpose of improving the management of coastal zones"
[257] Pallaoro, R. (2017). Value chain. Retrieved December 6, 2019, from Quizlet website: https://quizlet.com/it/253412570/value-chain-flash-cards/

different segments in the production cycle including a) inbound logistics, b) operations, c) outbound logistics, d) marketing & sales and e) service. The identification of different phases in the production cycle is important as segments do not have the same degree of (un)sustainability. The 'quick scan' VCA aims at identifying the generic opportunities and constraints regarding environmental, economic, social and governance sustainability for each segment, using a combination of methods such as a literature study, key interviews, focus groups and/or expert judgement.

Value chains consist of primary activities. These activities relate directly to the physical creation, sale, maintenance and support of a product or service. The five primary activities of each value chain include[258]:

- **Inbound logistics**: Refers to goods being obtained from the suppliers ready to be used for producing the end-product.
- **Operations**: The raw materials and goods obtained are manufactured into the final product. Value is added to the product at this stage as it moves through the production line.
- **Outbound logistics**: Once the products are manufactured, they are ready to be distributed to distribution centres, wholesalers, retailers or customers.
- **Marketing and Sales**: Marketing must make sure that the product is targeted towards the correct customer group. The marketing mix is used to establish an effective strategy; any competitive advantage is clearly communicated to the target group by the use of the promotional mix.
- **Services**: After the product/service has been sold, services relate to the support an organisation has to offer. This may come in the form of after sales training, guarantees and warranties.

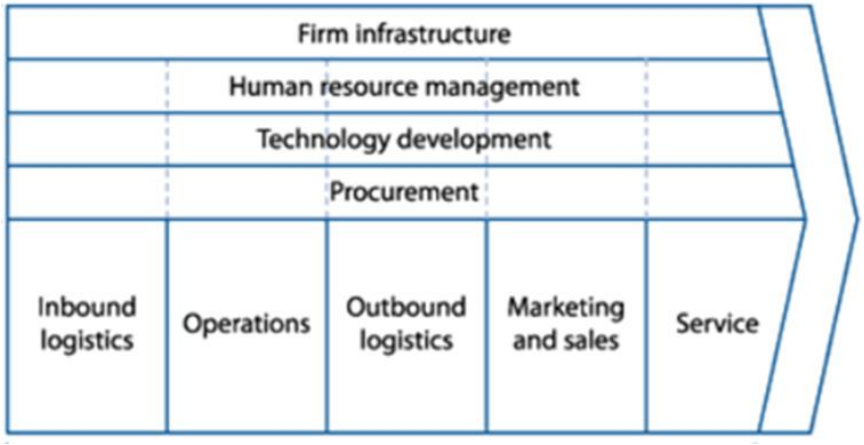
These primary functions are supported by a set of four supporting activities, namely[259]:

- **Firm infrastructure:** Every organisation needs to ensure that their finances, legal structure and management structure works efficiently and helps drive the organisation forward.
- **Human resource management:** The organisation will have to recruit, train and develop the correct people for the organization if they are to succeed in their objectives. Staff will have to be motivated and paid the 'market rate' if they are to stay with the organisation and add value to it over their duration of employment.
- **Technology development:** The use of technology to obtain a competitive advantage within the organisation. Technology can be used in production to reduce cost and thus add value, or in research and development to develop new products, or via the use of the internet so customers have access to online facilities.
- **Procurement**: This department must source raw materials for the organisation and obtain the best price for doing so. For the price they must obtain the best possible quality.

The original VCA theory of Porter focuses on the level of the firm or company. In the sections below the focus is primarily on the level of blue economy sectors.

[258] Porter, M. E. (1985). Value chain. The Value Chain and Competitive advantage: creating and sustaining superior performance.
[259] Porter, M. E. (1985). Value chain. The Value Chain and Competitive advantage: creating and sustaining superior performance..

Figure 4: Overview of primary activities and support activities for the VCA



Source: based on Porter, 1985

Based on the theoretical background as described above, the quick scan VCA is conducted using the two templates below (Table 4 and Table 5). For each of the case studies, a VCA was elaborated and contributed to the analysis of the activities (Task 5 and 6 of the study; the case studies are presented in Deliverable 4).

Table 4: Template to characterise primary activities of a blue economy sector

Segment	Short description of segment	Opportunities for sustainability	Constraints for sustainability
a. Inbound logistics			
b. Operations			
c. Outbound logistics			
d. Marketing and sales			
e. Service			

Table 5: Template to characterise support activities of a blue economy subsector

Segment	Short description
1. Sector infrastructure	
2. Human resource development	

Segment	Short description
3. Technology development	
4. Procurement	

Refinement of criteria, indicators, and units

Proposals for alternative and new indicators were first received from the experts conducting the research for the case studies where they found that the required data was either: i) not available at the level of detail required by the unit of the indicator or ii) not relevant for the sector addressed by the study. The suggested revisions for indicators were assessed by the study team and, if found suitable, included in the framework.

After this refinement process, further expert consultation was conducted via interviews and an online survey, to verify the acceptability of the indicators by sector representatives and other key stakeholders. Again, the suggestions made by the stakeholders were assessed and incorporated by the study team whenever possible. Feedbacks from EASME and DG MARE were integrated in the framework too. Annex 8 provides an overview of stakeholder engagement via a Peer Review Group, surveys, interviews and a Focus Group. An overview of the changes made to the BESF, with indications of where stakeholder feedback has been incorporated, is included in Annex 9.

The Blue Economy Sustainability Framework

The framework is composed of a set of criteria and indicators that measure the sustainability of a blue economy sector in terms of the four dimensions (environmental, economic, social and governance). These criteria (what/how to achieve) and indicators (what/how to measure) do not exist in isolation but underpin the working definition of a sustainable blue economy. In this study two types of criteria will be distinguished: common criteria and sector specific criteria. Indicators are the elements emerging from the criteria that should be measured. Each indicator is directly linked to a criterion; multiple indicators can be linked to a single criterion. The link between a criterion and the measured aspect of the blue economy, reported by the indicator, must be well understood. For each indicator there is a clear unambiguous technical definition of the data collected to calculate its value. Criteria and corresponding indicators are identified for the four dimensions of sustainability: environmental, economic, social and governance. The indicators are predominantly quantitative indicators, except for the governance dimension, for which qualitative indicators have been introduced as quantitative indicators “are notoriously difficult to measure”[260].

[260] http://www.sustain-eu.net/about_sustain/index.htm

Methodology for selecting the proposed key criteria and indicators

The selection of key criteria and indicators has been carried out based on the RACER methodology (Table 9). All common indicators have been scored by five independently working experts whose expertise covered all four dimensions of sustainability. The experts have used the matrix in Table 10 and Table 11for the scoring and selection of key criteria and indicators. Each indicator could be scored on a scale from 0 to 3.

Table 6: Overview of quality aspects used in the identification of key indicators

Quality aspect	Description	**Scoring**
Relevance	Indicators should be relevant for assessing sustainable blue economy	0 (not relevant) to 3 (highly relevant)
Acceptance	Indicators should be widely accepted by major stakeholders, academia and policy-making	0 (not accepted) to 3 (high level of acceptance)
Credibility	Indicators should be transparent, easy to interpret and understand by targeted audience	0 (not credible) to 3 (high level of credibility)
Easiness	Indicators should be based on available data in open databases	0 (no available data) to 3 (high availability of data)
Robustness	Indicators should be based on reliable data, updated regularly and consistently	0 (no reliable data) to 3 (highly reliable data)

The indicators with the highest scored have been ranked, selected and discussed in order to offer a balance in the four dimensions of sustainability. The list of 44 common indicators was reduced to a concise list of 20 key indicators.

Table 10: Matrix used for the identification of key criteria and indicators[261]

RACER Scoring of common indicators											
Relevancy: relevant for assessing sustainable Blue economy					**0**	**not relevant**					
Acceptance: widely accepted indicators by major stakeholders, academia and policy-making.					**1**	**low**					
Credibility: transparent, easy to interpret and understand by targeted audience.					**2**	**medium**					
Easiness: available data e.g. open databases, technical studies					**3**	**high**					
Robustness: reliable data, updated regularly and consistently.					15	Max score					
Common indicators							Expert x				
Environment											
	Code	**Criteria**	**Indicator**	**Unit**		**Type**	**R**	**A**	**C**	**E**	**R**	**Score**
1	C.EN.1	Air Quality	Exceedance of limit values for PM10/NOx/Sox	No. of days per year that limit values ar		Quantitative						
2	C.EN.2	Emissions to air	GHG emissions	Tonnes CO2 equivalent / year		Quantitative						
3	C.EN.3	Impact on ecosystems	Extent of habitat impacted	Area of impacted habitat in hectares		Quantitative						
4	C.EN.4	Level of energy consumption	Energy consumption	Tonnes of oil equivalent (TOE) /year		Quantitative						
5	C.EN.5	Level of energy consumption	Energy demand met by renewable energy	% total primary energy supply		Quantitative						
6	C.EN.6	Level of waste generated	Waste / marine litter generated	Tonnes of non-hazardous and hazardous waste generated/year		Quantitative						
7	C.EN.7	Waste management	Technology available for solid waste and waste water treatment	Yes/ No. If yes: specify		Qualitative						
											Total	
Economic / financial												
	Code	**Criteria**	**Indicator**	**Unit**		**Type**	**R**	**A**	**C**	**E**	**R**	**Score**
8	C.EC.1	Concentration of businesses	Existence of clusters	Yes/No		Qualitative						
9	C.EC.2	Economic viability	Gross value added (Size of the national / regional sector)	mEUR/year		Quantitative						
11	C.EC.3	Economic viability	Sector specific investments in the region	mEUR/year		Quantitative						
11	C.EC.4	Economic viability	Turnover	mEUR/year		Quantitative						
12	C.EC.5	Employment	Direct and indirect jobs	No. of direct and indirect jobs x1000 persons/year		Quantitative						
13	C.EC.6	Public funding	Public funding	% total income		Quantitative						
14	C.EC.7	Financial viability	Financial returns reinvested in local activities	% financial returns reinvested in local a		Quantitative						
15	C.EC.8	Economic benefits	Total revenues generated by local enterprises	% total revenues generated by local en		Quantitative						
16	C.EC.9	Financial viability	Financial self-sustainability of supported activities	Number of years required to achieve		Quantitative						
17	C.EC.10	Economic benefits	Local public revenue generated through time (taxes, fees, etc.)	mEUR/year		Quantitative						
18	C.EC.11	Financial viability	Additional streams of finance/investment attracted	mEUR/year		Quantitative						
											Total	

[261] Note that the RACER methodology was applied to a preliminary version of the criteria and indicator framework. The list of key criteria and indicators was further refined after expert consultation.

Table 11: Matrix used for the identification of key criteria and indicators (cont.)

	Code	Criteria	Indicator	Unit	Type	R	A	C	E	R	Score
Social											
19	C.SO.1	Education	Employees with post-school leaving age	% of employees with post-school leaving age educational qualifications	Quantitative						
20	C.SO.2	Employment conditions	Employees living below the national poverty levels	% employees living below the national poverty level	Quantitative						
21	C.SO.3	Employment conditions	Existence of 'a labour union'	Yes/ No	Quantitative						
22	C.SO.4	Employment conditions	Informal employment	% informal employment	Quantitative						
23	C.SO.5	Health and safety management	Fatal accidents and accidents at work	Accidents/ year	Quantitative						
24	C.SO.6	Health and safety management	Frequency of health and safety exercises and trainings	No. of emergency or safety drills	Quantitative						
25	C.SO.7	Inclusiveness	Employment rate of vulnerable groups (i.e. disabled persons, ethnic minority, etc)	% vulnerable people of total employees	Quantitative						
26	C.SO.8	Inclusiveness	Gender balance	% female employees	Quantitative						
27	C.SO.9	Level of acceptance by local community	Acceptance of environmental, economic and social impact by local communities	No. of reported actions of local communities against environmental, economic or social impacts	Quantitative						
										Total	
Governance											
	Code	Criteria	Indicator	Unit	Type	R	A	C	E	R	Score
28	C.GO.1	Certification and labelling	Use of certification or labelling	Yes/ No. If yes: specify	Qualitative						
29	C.GO.2	Impact assessment	Environmental Impact Assessment conducted and enforced via monitoring and evaluation	Scores: 1. no EIA conducted; 2. EIA conducted but not enforced; 3. EIA conducted and enforced via monitoring and evaluation	Qualitative						
30	C.GO.3	Innovation	Attention for innovation (or investment in R&D)	% revenue invested in R&D	Quantitative						
31	C.GO.4	Level of stakeholder engagement	Mechanism for stakeholder engagement	Scores: 1. Specific mechnism for stakeholders engagement besides public actors; 2. Occasional consultation with stakeholders, focused on public actors; 3. No stakeholders	Qualitative						
32	C.GO.5	Nature Based Solutions	Application of Nature Based Solutions	Scores: 1. Not applied; 2. Applied to some extend [example]; 3. Frequently applied [example]	Qualitative						
33	C.GO.6	Risk management	Existence / implementation of risk management plans	Scores: 1: No risk management plan; 2. Risk management plan exists; 3. Risk management plan exists and implemented	Qualitative						
34	C.GO.7	Spatial planning	Existence of a governmental plan that integrally approaches sectors to stimulate synergies and avoid conflicts (e.g. multi-use)	Yes / No	Qualitative						
35	C.GO.8	Strategy and vision	Reference to SDG goals in national / regional / local / company strategies	Yes/No. If yes: specify	Qualitative						
										Total	

Methodology for expert consultation

The final set of indicators developed for the blue economy sustainability criteria underwent an expert review for validation. The methodology for expert consultation consisted of the following steps:

1. Sampling experts
2. Drafting the questionnaire
3. Contacting experts
4. Gathering data and processing results
5. Conducting interviews

Sampling (step 1) was conducted by means of network sampling. The network sampling considered a proper distribution between the type of expert (network organization, industry, private company, consultant), blue economy sectors and geographical spread. A preliminary list was validated by the Client; after which the list was further improved. This resulted in a list of 32 experts.

1) The drafting of the questionnaire (step 2) was done by using the RACER principles (relevance, acceptance, credibility, easiness & robustness) as a starting point. This resulted in four questions which were applied to the set of sustainability indicators for a blue economy, both the common and the sector specific indicators. For each question, the expert could provide further input. For the overall framework, two questions were formulated (see tables below). The EU Survey Tool was used for the questionnaire.

Table 12: questions of the questionnaire used both for the common indicators and sector specific indicators. Multiple choice answers included "I overall agree", "I neither agree or disagree" and "I overall disagree".

* The indicators are suitable and complete for assessing the sustainability of the blue economy sector.
* The indicators are applicable to the blue economy sector.
* The indicators are transparent, easy to interpret or understandable.
* The indicators have reliable data that is (publicly) available.

Table 13: questions of the questionnaire used with respect to the overall framework. Multiple choice answers included "I overall agree", "I neither agree or disagree" and "I overall disagree".

* The indicators are suitable for assessing the sustainability of the sector.
* The set of indicators is complete and enables the assessment of sustainability of the sector.

The contacting of the experts (step 3) was conducted personally via the respective project team members, to increase the response rate. Out of the 32 experts contacted, 14 submitted a response to the questionnaire (45%).

The gathering of data and processing of results (step 4) is done via EU Survey. See Annex 8 for the results for questions concerning the common criteria/indicators and the full BESF. The results were organised into four categories 1) no action needed, 2) add to guidelines, 3) share comment with the parallel project "Unsustainable Finance in the Blue Economy", 4) modify criteria, indicators and/or units. The assessment illustrated that comments related to category 2 required clarifications of concepts such as "local/region" and "vulnerable groups". This has been integrated into the Guidelines as a glossary.

In order to discuss certain details in the feedback received, four out of the 14 experts who submitted responses via the questionnaire were invited for an interview. Interviews allowed the project team to further explain the purpose of the BESF and gave experts the opportunity to elaborate their suggestions. An overview of the outcomes of stakeholder engagement is provided in Annex 9.

Annex 8: Results of questionnaire for the validation of BESF indicators

A total of 14 experts submitted responses to the survey. Participants were able to choose for which subsector(s) they wanted to answer questions for. However, all respondents were asked to validate the common criteria/indicators of the BESF as well as the full framework (common indictors/criteria as well as the sector-specific indicators/criteria selected by the expert).

Table 15 provides and overview of the answers to questions concerning the overall validation and review of the reliability, applicability, credibility and easiness/ robustness of the **common criteria/indicators**.

Table 14: Questions on Common indicators

RACER questions on common indicators:		I overall agree	I neither agree nor disagree	I overall disagree
Relevance	The indicators are suitable and complete for assessing the sustainability of the blue economy sector.	5	6	3
Acceptance	The indicators are applicable to the blue economy sector.	9	4	1
Credibility	The indicators are transparent, easy to interpret or understandable.	7	6	1
Easiness/ Robustness	The indicators have reliable data that is (publicly) available.	7	6	1

Table 16 provides and overview of the answers to questions concerning the overall validation and review of the suitability and completeness of the **full framework** (common indictors/criteria as well as the sector-specific indicators/criteria selected by the expert).

Table 15: Questions on the full framework

Questions on full framework:	I overall agree	I neither agree nor disagree	I overall disagree
The indicators are suitable for assessing the sustainability of the sector.	4	6	4
The set of indicators is complete and enables the assessment of sustainability of the sector.	3	5	6

Annex 9: Overview of stakeholder engagement

Feedback from different stakeholders has fed into the development of the Blue Economy Sustainability Framework. The outcomes of discussions held in the Peer Review Group (17 March 2020), High-level Focus Group (22 July 2020) and via interviews with various industry representatives are presented below along with feedback obtained via the survey.

Table 16: Overview of stakeholder involvement in the development of the BESF

Sector	PRG	CS Expert	Survey	Interview	Focus Group	Outcome
Marine plant and algae harvesting; Aquaculture			x			No suggestions for improving criteria and/or indicators were done.
Marine plant and algae harvesting; Aquaculture			x			No suggestions for improving criteria and/or indicators were done.
Extraction of minerals; Renewable energy generation; Transport infrastructure; Coastal defence and flood protection			x			Based on this comment (and also a comment from the Client), C.GO.10 was added: “Existence of Life Cycle Assessment policy”.
Renewable energy generation; Tourism			x			Comments resulted in updating the guideline: induced impacts in tourism are not considered but are relevant. Also in the guideline the possibility to provide additional comments was added. An indicator was changed in order to not only focus on public funding: C.EC.11 on public/private funding. The unit of indicator STO.EC.1 (no. of tourists) was changed to no. of tourists/month in order to measure the seasonality.
Fish and shellfish harvesting; Marine plant and algae harvesting; Aquaculture			x			Comments resulted in a change of the unit of C.EN.2 (Emissions of CO2, SOx, NOx, and P.M.). Instead of measuring CO2 equivalents, CO2 is now separately measured. SFH.GO.1 was changed into “Multiannual management plans in place and implemented” in which fishing zones may be specified. SAQ.EC.5 (stocking density) was added.
Transport infrastructure; Transport shipping; Tourism; Coastal defence and flood protection			x	x		A representative from the organisation made suggestions in the survey and in the call that were considered by the project team. This resulted in changingC.SO.2 which now covers wage per gender
Extraction of minerals			x			The concept of “marine litter” was removed in C.EN.9 and made more generic by using “waste”. C.EN.1 was updated and now

Sector	PRG	CS Expert	Survey	Interview	Focus Group	Outcome
						includes “revenues invested in environmental causes” C.GO.10 was added to measure the existence of LCAs. For SEM.EN.1 the pressures of noise, vibration and heat were added.
Extraction of oil and gas			x	x		Suggestion to make the application to different scales/levels more explicit has been made processed in the final report and guideline. Further suggestions were made on checking frameworks such as Global Reporting Initiative, UN Global Compact, NBIM, Poseidon, EU Non Financial Reporting Directive, SDG14 by Fondation de la Mer. These were checked but no criteria/indicators were found with additional added value.
Renewable energy generation			x			Suggestion that impact on ecosystems can be both positive and negative was included in C.EN.3. The guideline has been updated to define the local level. C.EC.9 was added (Financial returns reinvested in local activities). C.EC.7 was updated in order to distinguish between direct and indirect jobs. C.GO.9 on existence of supply chain policy was added (also on request of Client).
Extraction of minerals			x			Feedback provided in email on top of survey contribution. C.EN.9 was updated to include waste recycling.
Extraction of minerals; Extraction of oil and gas; Fish and shellfish harvesting; Renewable energy generation			x			C.GO.2 on impact assessments now also includes Socio-Economic Assessments. SFH.GO.1 on fishery management plans was added to identify if/how fisheries are managed. It replaces the former indicator on discards.
Extraction of minerals; Extraction of oil and gas; Fish and shellfish harvesting; Renewable energy generation			x			Comments were similar to the ones of provided in the row above.
Transport infrastructure; Transport shipping			x	x		A representative from the organisation made suggestions in the survey and in the call. They related to specific types of shipping that were beyond the scope of the project. .
Other (Research Institute)	x					Indicators added on se/extension (e.g. oil spills, aquaculture and farms), Nature-based solutions could go under governmental instead of environmental, set of ‘baseline/key’ indicators should be established.
Other (NGO)	x					
Other (Intergovernmental Organisation)	x					
Other (Intergovernmental Organisation)				x		Provided suggestions on the use of the framework. No changes were made to criteria, indicators and guideline.

Sector	PRG	CS Expert	Survey	Interview	Focus Group	Outcome
Aquaculture		x				Reviewed the case study on shrimp farming and provided feedback. Emphasis was placed on the importance of VCA in order to reveal upstream/downstream activities that are important for sustainability.
Tourism				x		The organisation was invited to provide feedback via the survey but instead provided written comments via email. The interview involved explaining the process of the Blue Economy Lot 2 study, the outcome being that it should be made clear in the report that the indicators are preliminary. Additionally: - Indicator C.EN.2 on emission to air now includes types of emissions (also upon request of the Client). - Definition of local/regional added to the guidelines (also upon request of the Client). - C.EN.8 on energy efficiency added. - The guideline was updated to include relevant shipping indices and certification as part of C.GO.8.
Other (Business association)						The organisation were invited to provide feedback via the survey but instead provided written comments via email. This did not lead to any changes in the criteria, indicators or guideline.
Other (Intergovernmental Organisation)					x	Comments from discussion will feed into the recommendation of the report. It should be made clear in the report what the next steps should be in order to bring the BESF to a higher level. In doing so, we should emphasize the importance of our framework as a preliminary step. After the focus group meeting, an NGO provided input by email. This lead to the following changes: SFH.EN.2 (selective gear) and SFH.EN.3 (non-destructive gear) were added (also upon request of the Client). - SRE.EN.1 changed into “Species fatalities due to collisions”. C.EC.13 on maintenance costs added. - STO.GO.3 on sustainable sourcing added (also upon request of the Client). - STS.EN.1 added on exhaust cleaning.
Other (Intergovernmental Organisation)					x	
Other (European Institution)					x	
Other (NGO)					x	
Fish and shellfish processing			x			Provided feedback via survey. Two indicators were added:

Sector	PRG	CS Expert	Survey	Interview	Focus Group	Outcome
						- SFP.EC.4: Domestic fish production vs. imports - SFP.EC.5: Type of imports (also upon request of the Client).

Annex 10: Comprehensive database of blue economy criteria and indicators

The table below provides an overview of the development of the Blue Economy Sustainability Framework (BESF). The criteria, indicators and units of the BESF have been modified in several stages (with desk further research taking place in parallel):

1. The initial indicators (as presented in the interim report for this study) represent the common indicators as proposed after initial desk research was conducted.
2. Following this, the case studies (CS) applied indicators that had been developed for a specific subsector. These were used in addition to the common indicators.
3. In the process of applying these indicators in the case studies, new or alternative criteria/indicators/units were suggested[262].
4. Intermediate criteria/indicators/units were presented to members of the Peer Review Group (PRG) on 17 March 2020.
5. New or alternative criteria/indicators/units/domains were suggested by the Peer Review Group (PRG).
6. Furthermore, the European Commission (DG-MARE /EASME) have provided feedback on the proposed criteria and indicators at four separate stages throughout the project[263].
7. Following the High-level Focus Group (FG) taking place on 22 July 2020, feedback was received via an email from one of the participants
8. A survey was distributed (on the 6 June 2020), inviting sector representatives to review the BESF. Following this, a series of interviews were organised with experts from the sectors. Some of the feedback received has been incorporated into the BESF. The last input for the survey was received on the 17 August 2020.
9. The final stage of the table below represents the proposed criteria/indicators/units of the BESF as presented in the final report.

Indicators and criteria are either common for all sectors (**C**), or specific to one of the following subsectors: Extraction of Minerals (**SEM**), Extraction of oil and gas (**SEOG**), Desalination (**SDE**), Coastal defence and flood protection (**SCD**), Fish and shellfish harvesting (**SFH**), Aquaculture (**SAQ**), Fish and shellfish processing (**SFP**), Marine plant and algae harvesting (**SPH**), Renewable Energy (**SRE**), Tourism (**STO**), Transport infrastructure (**STI**) and Transport Shipping (**STS**).

Table 17: Overview of the development of the BESF

[262] Indicators marked with an 'x' in the column 'Suggested by experts' represents indicators that were suggested by the study team after applying a Value Chain Analysis.
[263] Upon the (re)submissions of the (draft)final report on the 20th March, 2nd July, 7th August and 15th September 2020.

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
Ec.	C	Economic viability	Investment in specific sector over time compared to average investment in the region	Mln EUR	x	x
Ec.	C	Public support in the form of public funding	Public funding	m EUR/year	x	x
Ec.	C	Economic viability	Turnover	Million EUR/year	x	x
Ec.	C	Economic viability	Gross value added (Size of the national / regional sector)	Million EUR/year	x	x
Ec.	C	Concentration of businesses	Existence of clusters	Yes/ No	x	x
Ec.	C	Employment	Direct and indirect jobs	No. of direct and indirect jobs per 1000 persons/year	x	x
Ec.	C	Economic benefits	Local public revenue generated through time (taxes, fees, etc.)	mEUR/year		
Ec.	C	Economic benefits	Total revenues generated by local enterprises	% total revenues generated by local enterprises		
Ec.	C	Economic viability	Sector specific investments in the region	Million EUR/year		
Ec.	C	Financial viability	Additional streams of finance/investment attracted	m EUR/year		
Ec.	C	Financial viability	Financial returns reinvested in local activities	% financial returns reinvested in local activities		
Ec.	C	Financial viability	Financial self-sustainability of supported activities	Number of years required to achieve the full financial self-sustainability of supported activities (e.g. debt-to-equity ratio)		
Ec.	C	Public funding	Public funding	% total income		
Ec.	C	Funding	Public / private funding	% of turnover		
Ec.	C	Costs	Average personnel costs	x1000 EUR / year		
Ec.	C	Costs	Maintenance costs	mEUR/year		
En.	C	Water management	Water quality	Quantity of hydrocarbons and/or other / year	x	x
En.	C	Pollution	Waste / marine litter	Tonnes of waste / marine litter / year	x	
En.	C	Air Quality	Exceedance of limit values for PM10/NOx/Sox	No. of days per year that limit values are exceeded for PM10/NOx/Sox / year	x	x
En.	C	Level of waste generated	Waste / marine litter generated	Tonnes of non-hazardous and hazardous waste generated/year.		x
En.	C	Positive/negative Impact on ecosystems	Extent of habitat positively/negatively impacted	Area of positively/negatively impacted habitat in hectares, Number of species impacted		x
En.	C	Level of energy consumption	Energy consumption	Tonnes of oil equivalent (TOE) /year	x	x

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
En.	C	Nature Based Solutions	Application of Nature Based Solutions	Score 1. Not applied 2. Applied to some extend [example] 3. Frequently applied [example]		x
En.	C	Emissions to air	Emissions of CO2, SOx, NOx, and P.M.	Tonnes of CO2 equivalent / year Tonnes of SO2 equivalents / year Tonnes of NO2 equivalents / year Tonnes of pollutant / year	x	x
En.	C	Level of energy consumption	Energy demand met by renewable energy	% total primary energy supply		x
En.	C	Waste management	Technology available for solid waste and waste water treatment	Yes/ No. If yes: specify		x
En.	C	Mitigation	Gross value or percentage of revenue invested in environmental causes related to the sector's activities directly (e.g. mitigation, restauration, monitoring) or indirectly (offsetting).	mEUR/year or % of revenue/year		
En.	C	Impact on ecosystems	Extent of coastal and marine habitat positively/negatively impacted	Area of positively and negatively impacted habitat in hectares		
En.	C	Impact on ecosystems	Threatened species (IUCN red list) of known species	%		
En.	C	Waste / water management	Waste generated and recycled; Waste-water generated and reused	Tonnes of waste generated and recycled /year. Million m3 of waste-water generated and reused/year.		
En.	C	Impact on ecosystems	Support given to local entities working on the protection, conservation and management of local biodiversity and landscapes	% of turnover dedicated to such support or, f in-kind support (such as making manpower or machinery available free of charge, or donating land), specify		
En.	C	Energy efficiency	Measures taken to increase energy efficiency	Yes / no. If yes, specify		
G	C	Impact Assessment	Environmental Impact Assessment conducted and enforced via monitoring and evaluation	Scores 1. no EIA conducted, 2. EIA conducted but not enforced 3. EIA conducted and enforced via monitoring and evaluation	x	x

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
G	C	Involvement of local / regional administration	Local / regional	Involvement of local / regional administration	x	x
G	C	Level of stakeholder engagement	Mechanism for stakeholder engagement	Score 1. Specific mechanism for stakeholders engagement besides public actors 2. Occasional consultation with stakeholders, focused on public actors 3. No stakeholder involvement	x	x
G	C	Strategy and vision	Reference to SDG goals in national / regional / local / company strategies	Yes/ No	x	x
G	C	Spatial Planning	Existence of a governmental plan that integrally approaches sectors to stimulate synergies and avoid conflicts (e.g. multi-use)	Yes/ No. If yes: specify if/how the activity is integrated in the plan	x	x
G	C	Certification and labelling	Use of certification or labelling	Yes/ No. If yes: specify		x
G	C	Certification and labelling	Existence of a sustainability label or certificate	Score 1: No sustainability label or certification 2. Sustainability label(s) or certification exists/awarded (please specify) 3. Sustainability label(s) or certification applied		
G	C	Risk management	Existence / implementation of risk management plans	Score 1: No risk management plan 2. Risk management plan exists 3. Risk management plan exists and implemented		
G	C	Innovation	Attention to innovation (or investment in R&D)	% revenue invested in R&D		
G	C	Nature Based Solutions	Application of Nature Based Solutions	Score 1. Relevant, but not applied 2. Applied to some extent [example] 3. Frequently applied [example] 4. Not applicable to the company/sector activities		
G	C	Impact Assessment	Environmental Impact Assessment (EIA), Strategic Environmental	Scores 1. no EIA/SEnA/SEcA		

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
			Assessment (SEnA) and Socio-Economic Assessment (SEcA) conducted and enforced via monitoring and evaluation	conducted, 2. EIA/SEnA/SEcA conducted but not implemented/enforced 3. EIA/SEnA/SEcA conducted and enforced via monitoring and evaluation		
G	C	Supply chain	Existence of supply chain policy	Yes/No. If yes: specify		
G	C	Permits	Typical permitting regime followed prior to operations	Score: 1: Permit with EIA procedure; 2: Permit procedure required, but below EIA threshold; 3: No permitting or environmental administration required		
G	C	Risk management	Existence / implementation of risk management plans taking into account the precautionary principle	Score 1: No risk management plan 2. Risk management plan exists 3. Risk management plan exists, includes precautionary principles and is implemented		
G	C	Strategy and vision	Integration of SDGs in the company's strategy and operations	% of activities covered by SDG reporting		
G	C	Climate change	Measures taken for climate change adaptation	Yes/No. If yes: specify		
G	C	Education on sustainability	Participation in information and training sessions about sustainability	Yes/ No, if yes specify		
G	C	Supply chain	Existence of Life Cycle Assessment policy	Yes/No. If yes: specify		
S	C	Health	Access to health care	% of employees with access to healthcare	x	
S	C	Communities	Impact on local communities	Amenities, infrastructure	x	x
S	C	Employment conditions	Employees living below the national poverty levels	% employees living below the national poverty level	x	x
S	C	Inclusiveness	Employees with no post-school diploma?	% of employees with no post-school diploma?	x	
S	C	Health and safety management	Fatal accidents and accidents at work	Accidents/ year		
S	C	Inclusiveness	Gender balance	% female employees	x	x
S	C	Employment conditions	Existence 'a labour union'	Yes/ No		
S	C	Employment conditions	Presence and level of activity of labour unions in the company/sector in the company/sector	Yes/No. If yes, specify		
S	C	Employment conditions	Informal employment	% informal employment put of total employment		

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
S	C	Health and safety	Safety/security	No. of accidents at work		x
S	C	Health and safety management	Fatal accidents and accidents at work	Lost Time Incidents / year		
S	C	Health and safety management	Frequency of health and safety exercises and trainings	No. of emergency or safety drills to prevent fatal accidents and accidents at work		x
S	C	Health and safety management	Existence of policies and measures to combat occupational diseases and accidents	Yes/no, if yes: specify		
S	C	Fairness in remuneration	Evidence of unequal pay between social categories for equal work	Yes/no, if yes: explain evidence, type of work and social categories affected, degree of discrimination in pay		
S	C	Inclusiveness	Employment rate of vulnerable groups (i.e. disabled persons, ethnic minority, etc)	% vulnerable people of total employees		x
S	C	Level of acceptance by local community	Acceptance of environmental, economic and social impact by local communities	No. of reported actions of local communities against environmental, economic or social impacts		
S	C	Employment conditions	Average wage of employees compared to sector average or national average	EUR/year, divided in male/female/other		
S	C	Inclusiveness	Employment rate of vulnerable groups	% vulnerable people of total work force per social category (*see guidelines).* For every social category define: •Gender (% male/female/other) •Average age		
S	C	Health and safety management	Frequency of auditing by external health & safety experts	No. of audits by external health and safety experts, including evidence of application in practice such as technical measures, regular medical screenings, etc.		
Ec.	SAQ	Feed management	Realised Feed Conversion Ratio	Feed Conversion Ratio		
Ec.	SAQ	Economic viability	Average size of farm	Hectares of land		
Ec.	SAQ	Economic viability	Production 'farmed fish' (monetary)	m EUR/year		
Ec.	SAQ	Economic viability	Production 'farmed fish' (weight)	Tons/year		
Ec.	SAQ	Economic viability	Average size of farm	Hectares of land or water		

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
Ec.	SAQ	Economic Viability	Stocking density	No. of fry/m2		
En.	SAQ	Parasites	Aquaculture farms with lice threshold above national threshold	No. of farms		
En.	SAQ	Chemical use	On-farm documentation available with detailed information on chemicals use, compliant with regulations (including anti-biotics)	Yes/ No. If yes: specify		x
En.	SAQ	Feed management	Responsible origin of raw materials in feed	Yes/ No. If yes: specify		x
En.	SAQ	Farm management	Number of escape events	Number of escaped events / year		
En.	SAQ	Farm management	Number of escaped fish	Number of escaped fish / year		
En.	SAQ	Farm management	Mortalities reduction program exists and implemented	Yes/ No. If yes: specify		x
En.	SAQ	Water quality	Measures taken to reduce nutrient emissions	Yes/ No. If yes: specify		
En.	SAQ	Water quality	Phosphorous (P) and nitrogen (N) concentration	mg/L		
En.	SAQ	Supply chain	Existence and effective implementation of a company policy to ensure inputs/raw materials are obtained from sustainable sources	Score 1. Policy does not exist 2. Policy exists but not implemented 3. Policy exists and implemented		
En.	SAQ	Impact on ecosystems	Refuge effect for species	Yes/no, if yes: specify		
Ec.	SCD	Durability of structure	Estimated lifespan of the structure	No. of years		
Ec.	SCD	Costs	Maintenance costs	Million EUR/year		
Ec.	SCD	Costs	Total construction costs	Million EUR		
Ec.	SCD	Flood safety	Target flood protection level	1 failure in x number of years		
Ec.	SCD	Durability of structure	Age of the structure	No. of years		
En.	SCD	Flood safety	Open vs. closed coastal defence/flood protection system	Open or closed system. If closed, specify impact on fish migration.		
En.	SCD	Flood safety	Existence of natural barriers	No. and type of natural barrier (e.g. (wetlands, mangroves, reefs)		
En.	SCD	Impact on ecosystems	Refuge effect for species	Yes/no, if yes: specify		
Ec.	SDE	Infrastructure capacity	Volume of extracted seawater	m3/day		x
Ec.	SDE	Infrastructure capacity	Fresh water produced	m3/day		x

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
Ec.	SDE	Infrastructure capacity	Quality of water produced	mg TDS/L		x
Ec.	SDE	Level of energy consumption	Energy used per unit fresh water produced	kWh / m3		x
Ec.	SDE	Economic viability	Levelized cost of water production	EUR/m3 product water		x
En.	SDE	Impact on ecosystems	Salinity increase	ppm above ambient salinity		
En.	SDE	Impact on ecosystems	Temperature increase	°C above ambient temperature		
En.	SDE	Infrastructure capacity	Amount of discharged brine	MT/year		
En.	SDE	Chemical use	Discharge of chemicals.	Tons/chemical/year		
G	SDE	Infrastructure capacity	Measures taken for strategic buffering in periods when water demand is low.	Yes /no. If yes specify		
G	SDE	Innovation	Reuse of brine, i.e. as source of valuable raw material	Yes /no. If yes specify		
Ec.	SEM	Economic viability	Production 'marine aggregates' (weight)	Million tons / year		
Ec.	SEM	Economic viability	Value 'marine aggregates' (monetary)	Million EUR/year		
En.	SEM	Impact on environment	Technologies applied to reduce dredging plume / turbidity	No. and type of technologies		x
En.	SEM	Impact on environment	Number of times that turbidity is not in compliance with regulations	No./year		
En.	SEM	Impact on environment	Technologies applied to reduce the impact of dredging plume, noise, vibration and heat	No. and type of technologies		
Ec.	SEOG	Economic viability	Levelized cost of energy production	Tons of oil equivalent/year		
Ec.	SEOG	Economic viability	Production 'oil and gas' (monetary)	Million EUR/year		
Ec.	SEOG	Economic viability	Production of oil and gas (tons)	Tons of oil equivalent / year		
En.	SEOG	Emissions to water	Produced water subject to treatment	% produced water subject to treatment		
En.	SEOG	Oil spill response	Frequency of Oil Spill Response exercises and trainings	No. of exercises and trainings / year		
En.	SEOG	Waste management	Existence of drilling waste management plan	Yes/ No		
En.	SEOG	Impact on environment	Technologies applied to reduce the impact of noise, vibration and heat	No. and type of technologies		
En.	SEOG	Impact on ecosystems	Refuge effect for species	Yes/no, if yes: specify		
Ec.	SFH	Economic viability	Production 'harvested fish and shellfish' (monetary)	Million EUR/year		
Ec.	SFH	Economic viability	Production of harvested fish and shellfish (weight)	Landings weight in tons / year		

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
En.	SFH	Status of stock	Exploitation of stock at Maximum Sustainable Yield	% stock exploited at Maximum Sustainable Yield (per species)		
En.	SFH	Fishing gear	Use of non-destructive fishing techniques/gears	Yes / no. If no, specify		
En.	SFH	Fishing gear	Use of selective fishing techniques/gears	Yes / no. If no, specify		
G	SFH	Fishery management	Discard regulations in place and enforced	Yes/ No		
G	SFH	Fishery management	National Plan of Action for Illegal, Unregulated, Unreported Landings	Yes/ No, if yes specify		
G	SFH	Fishery management	Quota system in place and implemented	Yes/ No, if yes specify		
G	SFH	Fishery management	Multiannual management plans in place and implemented	Yes/ No, if yes specify		
G	SFH	Fishery management	Fishing vessels equipped with electronic positioning and catch reporting device	% of the unit of analysis		
Ec.	SFP	Economic viability	Production 'processed fish and shellfish' (monetary)	Million EUR/year		
Ec.	SFP	Economic viability	Production of processed fish and shellfish (weight)	Tons / year		
Ec.	SFP	Processing conditions	Access to ice / refrigeration	Yes/ No		
Ec.	SFP	Economic viability	Domestic fish production vs. imports	Tons/year		
Ec.	SFP	Economic viability	Type of imports	Tons/year for the following categories: - (Frozen) whole fish - (Frozen) fillet and steak - Canned fish and shellfish - Dried (frozen) fish - Fresh fish and shellfish		
En.	SFP	Waste management	Re-use of fish waste	Yes/ No		
En.	SFP	Waste management	Use of recycled packaging materials	Yes/ No		
En.	SFP	Wastewater management	Treatment of waste water	Yes/ No		
S	SFP	Social balance	Effect of fish input purchases on: • local prices • local harvesters • users of fish	Yes/no, if yes specify for each category		
G	SFP	Supply chain	Existence and effective implementation of a company policy to ensure inputs/raw materials are obtained from sustainable sources	Score 1. Policy does not exist 2. Policy exists but not implemented		

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
				3. Policy exists and implemented		
Ec.	SPH	Economic viability	Production 'harvested marine plants and algae' (monetary)	Million EUR/year		
Ec.	SPH	Economic viability	Production of harvested marine plants and algae (weight)	Landings weight in tons / year		
En.	SPH	Status of stock	Exploitation of stock at Maximum Sustainable Yield	% stock exploited at Maximum Sustainable Yield (per species)		
En.	SPH	Harvesting management	Use of selective harvesting techniques/gears	Yes / no. If no, specify		
En.	SPH	Harvesting management	Use of non-destructive harvesting techniques/gears	Yes / no. If no, specify		
G	SPH	Harvesting management	National Plan of Action for Illegal, Unregulated, Unreported Landings	Yes/ No		
G	SPH	Harvesting management	Quota system in place	Yes/ No		
G	SPH	Harvesting management	Existence and enforcement of a stock management policy	Score 1. Policy does not exist 2. Policy exists but not enforced 3. Policy exists and enforced		
G	SPH	Harvesting management	Monitoring and inspection of harvesting efforts	Yes/no, if yes specify		
Ec.	SRE	Economic viability	Distance from shore	Km		
Ec.	SRE	Infrastructure capacity	Electrical capacity	MW		
Ec.	SRE	Infrastructure capacity	Installed capacity relative to surface used	MWh/Km2		
Ec.	SRE	Economic viability	Levelized cost of energy production	Euro/MWh		
Ec.	SRE	Infrastructure capacity	Total gross electricity generation	MWh/year		
En.	SRE	Impact on ecosystems	Species fatalities due to collisions	No. and type of specie fatalities/year		
En.	SRE	Impact on ecosystems	Refuge effect for species	Yes/no, if yes: specify		
Ec.	STI	Economic viability	Production 'cargo capacity'	In Twenty Foot Equivalent Unit (TEU)/year		
Ec.	STI	Economic viability	Production 'cargo capacity' (monetary)	Million EUR/year		
Ec.	STI	Flood safety	Measures taken to ensure flood safety	Yes/ No. If yes: specify		x
En.	STI	Introduction of invasive species	Onboard ballast water treatment system available and functioning	Yes/ No		x

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
En.	STI	Oil spill response	Frequency of Oil Spill Response exercises and trainings	No. of exercises or trainings / year		x
En.	STI	Water quality	Measures taken to reduce nutrient emissions	Yes/ No. If yes: specify		x
En.	STI	Use of shore power	Availability of shore power infrastructure in port	Yes/ No		
G	STI	Sustainable infrastructure	Measures taken to ensure reliable, sustainable and resilient infrastructure	Yes/ No. If yes: specify		
G	STI	Hazardous waste management	Guidelines and plans for handling hazardous substances	Yes / no. If yes: specify		
Ec.	STO	Economic viability	Number of tourists	No. of tourists/year		
Ec.	STO	Economic viability	Number of tourists	No. of tourists/month		
En.	STO	Level of energy consumption	Specific energy use	kWh/m2yr		
En.	STO	Level of water consumption	Water consumption per guest night	Litres/guest night		
G	STO	Development control	Existence of land use or development planning processes, including tourism	(%) the area subject to planning management		
G	STO	Development control	Existence of visitor taxes and fees with the aim of re-investing in mitigating or reversing negative effects on the local ecosystems and community	Yes/no, if yes specify		
G	STO	Supply chain	Existence and effective implementation of a company policy to ensure inputs/raw materials are obtained from sustainable sources	Score 1. Policy does not exist 2. Policy exists but not implemented 3. Policy exists and implemented		
S	STO	Employment conditions	Seasonal jobs	% jobs that are seasonal		
S	STO	Social balance	Tourism density	Ratio of total number of nights spent relative to the total surface area of the region		
S	STO	Social balance	Tourism intensity	Ratio of nights spent at tourist accommodation establishments relative to the total permanent resident population of the area		
Ec.	STS	Economic viability	Production 'cargo capacity'	Twenty Foot Equivalent Unit (TEU) /year		
Ec.	STS	Economic viability	Cargo tonnage	Million EUR/year		

Dimension	Sector	Criteria	Indicator	Unit	Initial Indicators	Applied to CS
En.	STS	Introduction of invasive species	Onboard ballast water treatment system available and functioning	Yes/ No		
En.	STS	Emissions to air	Measures taken to reduce emissions to air	Yes/ No. If yes: specify		
En.	STS	Emissions to air	Use of low Sulphur Heavy Fuel Oil	Yes/ No		
En.	STS	Waste management	Waste management systems (sludge handling) available and functioning	Yes/ No		
En.	STS	Level of fuel consumption	Fuel consumption	Tons/kind of fuel/year		
En.	STS	Impact on ecosystems	Sewage discharge in Particular Sensitive Sea Areas	Yes / no		
En.	STS	Chemical use	Use of chemicals for antifouling, stern tube oils, external hydraulic fluids, gear oils for thrusters and controllable pitch propellers, boiler/cooling water treatment, cleaning agents, refrigerants.	Yes/no for each application. If yes: specify.		
En.	STS	Use of shore power	Onboard infrastructure to connect to shore power	Yes/ No		
En.	STS	Emissions to air	Measures taken to reduce emissions to air through exhaust cleaning	Yes/ No. If yes: specify if measure involves disposal of sludge produced by Exhaust Gas Cleaning System		
En.	STS	Emissions to air	Average fuel Sulphur content per bunkering	% avg, fuel Sulphur content per kind of fuel		

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